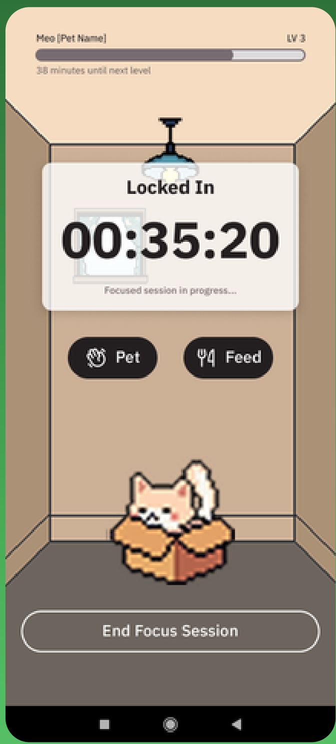
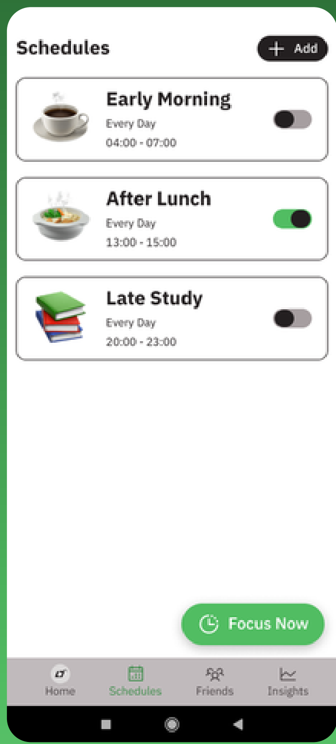
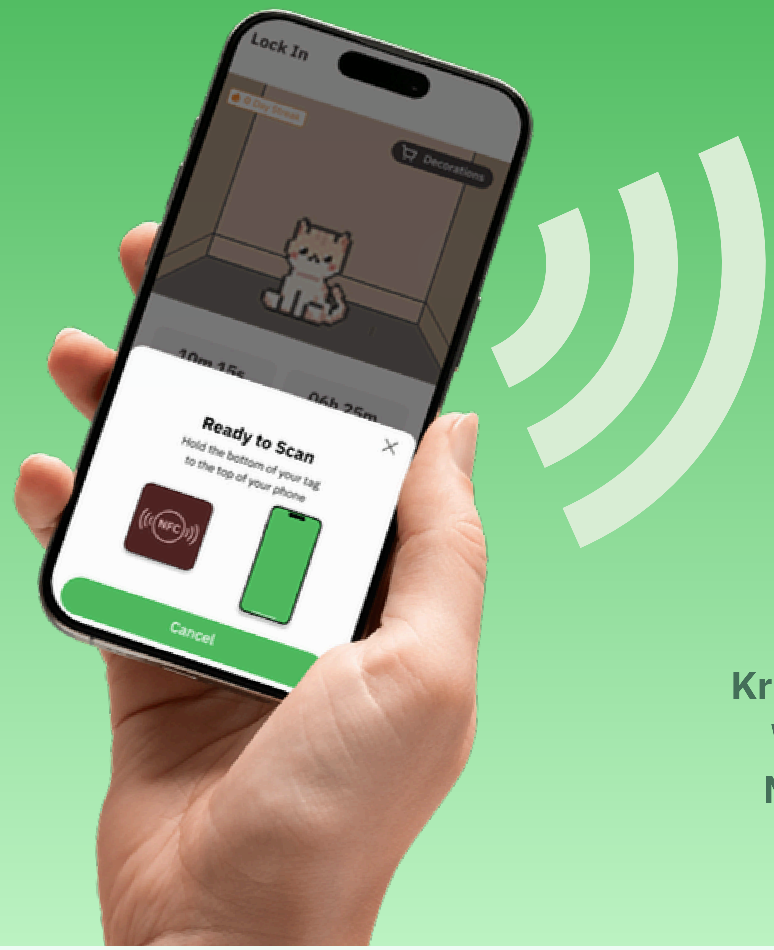
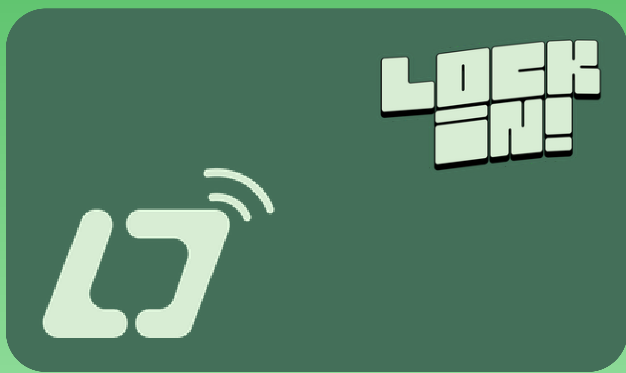
LOCK-IN

Mitigating Attention-Span Issues in Adolescent and Young-Adult Generations through a Physically Activated Focus Application









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Engineering Project Final Report

Lock-In: Mitigating Attention-Span Issues in Adolescent and Young-Adult Generations through a Physically Activated Focus Application

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Abstract

Excessive smartphone use among adolescents and young adults has become a significant public-health concern, contributing to chronic attention fragmentation, sleep disruption, and reduced productivity. While focus and productivity applications exist to mitigate these effects, they share a critical design flaw: relying entirely on software-based activation, they can be bypassed instantaneously, and the absence of sustained motivational mechanisms leads to abandonment rates exceeding 70 percent within the first week. This project presents **Lock-In**, a gamified focus app-blocker application designed to address both failure modes through the integration of physical activation and positive-reinforcement gamification.

Lock-In requires users to tap an NFC-enabled external device to initiate a focus session, introducing a deliberate physical friction that substantially raises the cost of impulsive bypass. During active sessions, the application blocks access to user-configured distraction applications and optionally routes unlock attempts through a micro-habit inducer, prompting a brief constructive action before granting temporary access. Continued focus time is rewarded through a virtual companion pet system, in which the companion's growth and customizable room environment are tied directly to accumulated focus hours, providing ongoing intrinsic motivation. A statistics dashboard and social leaderboard complement these features by supporting personal progress tracking and peer accountability.

The system is implemented as a Flutter/Dart Android application backed by a RESTful API and cloud database. Its usability and user experience are evaluated through a structured User Acceptance Testing protocol comprising ten task-based scenarios administered to five to eight representative participants, with quantitative metrics including Task Success Rate, Time on Task, Single Ease Question (SEQ) scores, and System Usability Scale (SUS) scores, supplemented by qualitative think-aloud data and semi-structured debrief interviews. The project demonstrates a feasible and user-aligned approach to mitigating smartphone distraction by combining behavioral science, physical activation, and gamified habit formation.

Table of Contents

List of Figures

List of Tables

1. Introduction

Smartphone usage has become an inescapable feature of contemporary life. While these devices offer unparalleled convenience for communication, education, and entertainment, their unconstrained use has given rise to a growing public-health concern: attention fragmentation and compulsive engagement, particularly among adolescents and young adults. Studies indicate that nearly one in five adolescents spend the majority of their leisure time on smartphones [1], and surveys conducted for this project reveal that the typical young adult now spends seven to nine hours per day on their device, of which approximately 55 percent is self-assessed as unproductive. At a systemic level, such patterns threaten cognitive development, reduce academic and professional output, and erode sleep quality and general well-being [2].

The root causes of this phenomenon are well-documented. Social media platforms and mobile games are deliberately engineered to exploit dopamine-driven reward circuits, using mechanisms such as infinite scroll, push notifications, and unpredictable social feedback to sustain compulsive checking behavior [3]. Critically, the problem is compounded by the inadequacy of existing countermeasures. Current productivity and focus applications, while conceptually sound, are undermined by a fundamental architectural weakness: they rely exclusively on software-based activation and deactivation. Because the lock and unlock controls reside on the same device they are meant to protect, determined or impulsive users can circumvent them within seconds, negating any therapeutic or behavioral benefit.

A second failure mode of existing tools is motivational: they impose restrictions without providing meaningful, sustained incentives for continued use. Industry data suggests that more than 50 percent of productivity-app users disengage within the first 100 days, and general behavioral-app analytics indicate that as many as 70 percent of users abandon an application within its first week [6,7]. This churn rate reflects a design gap between the clinical rationale for focus tools and the behavioral reality of their users.

This project, Lock-In, addresses both failure modes through the development of a gamified focus-blocker mobile application for Android that introduces physical activation as a security measure and positive-reinforcement gamification as a retention mechanism. The physical activation component requires the user to tap an NFC-enabled external device (such as a wristband, keychain, or card) to initiate and validate a focus session, introducing a deliberate friction that substantially raises the cost of bypassing the application. The gamification layer replaces the punitive tone of conventional blockers with a virtual companion pet whose growth, health, and customizable environment are tied directly to the user's accumulated focus time, creating an intrinsic reward structure aligned with productive behavior. Complementary features include a micro-habit inducer, a statistics dashboard, and a social leaderboard, all designed to sustain engagement and encourage the gradual formation of healthier digital habits.

A core design constraint that distinguishes Lock-In from conventional gamified applications is the principle of sustainable engagement. The companion system is engineered so that the application rewards time spent away from the phone rather than time spent inside it. The companion's growth mechanics are tied exclusively to cumulative focus outcomes; its in-app interaction surface is deliberately minimal; and no streak-loss or decay penalties are imposed that could shift user motivation from genuine focus achievement toward anxious score-protection. The explicit goal is a system that fades into the background once a session begins, providing a meaningful reward when the session ends — not a new source of habitual app-checking.

The primary objectives of this project are:

1. Develop a reliable app-blocking mechanism that prevents user access to blacklisted applications during active focus sessions.
2. Implement a physical NFC-based activation system that eliminates software-only bypass vulnerabilities and reduces impulsive override.
3. Design and integrate a micro-habit inducer that requires users to perform brief positive actions as the condition for unlocking phone access during focus time.
4. Build a virtual companion pet system that provides sustained, intrinsic positive reinforcement tied to cumulative focus achievement.
5. Create a statistics and social leaderboard dashboard that enables users to monitor personal progress and engage in community-based accountability.
6. Evaluate the usability, learnability, and user experience of the application through a structured UAT with representative participants.

The remainder of this report is organized as follows. Section 2 reviews the academic and empirical literature underpinning the problem and solution strategy. Section 3 details the design requirements. Section 4 describes the design and implementation of each system component. Section 5 presents the experimental setup and evaluation methodology. Section 6 reports and discusses the results. Section 7 concludes the report. Sections 8 and 9 provide references and appendices, respectively.

2. Background

This section synthesizes the academic and empirical foundations underlying the Lock-In project. It examines the neurobiological mechanisms driving smartphone addiction, the documented consequences of uncontrolled use, the limitations of current intervention strategies, and the evidence base for the gamification and physical-activation approaches adopted in this project.

2.1 Neurobiological and Behavioral Mechanisms of Smartphone Addiction

Smartphone addiction is a recognized behavioral phenomenon that shares structural similarities with substance addiction, rooted in the brain's dopaminergic reward circuit [1]. Each notification, message, or social-media interaction delivers an unpredictable micro-reward that triggers dopamine release, reinforcing the impulse to check the device repeatedly. This mechanism is deliberately amplified by platform designers who employ variable-ratio reinforcement schedules to maximize checking frequency [3]. Figure 1 illustrates this feedback loop, showing how a stimulus (e.g., a notification) triggers a dopamine spike that reinforces the compulsive checking behavior, creating a self-sustaining cycle.

Figure 1. Dopamine Feedback Loop Diagram

The dual-pathway theory of addiction provides a useful framework for understanding the trajectory of problematic smartphone use [3]. In the early phase, use is driven by positive reinforcement: the pursuit of pleasurable social rewards such as likes, comments, and messages. Over time, the reinforcement mechanism shifts to negative reinforcement: using the device to escape stress, anxiety, or boredom. This second pathway is empirically shown to exert a stronger driving force on sustained addiction than perceived enjoyment alone [4]. The implication for intervention design is significant: effective tools must address both pathways, offering alternative sources of reward while raising the barrier to impulsive escape use.

The prefrontal cortex, responsible for impulse control and executive function, is progressively impaired by chronic dopamine over-stimulation. Users become less capable of resisting the urge to check their device even when they are consciously aware that doing so is counterproductive. This neurological reality explains why software-only interventions, which rely entirely on the user's willingness not to override them, are inherently limited in efficacy.

2.2 Consequences of Uncontrolled Smartphone Use

From a cognitive and productivity standpoint, compulsive checking behavior acts as repeated interruptions that prevent users from achieving a state of sustained attention or flow [2]. Research indicates that it may take students up to 20 minutes to regain focus on a learning task after a single smartphone-related interruption [5]. Sleep disruption is a further well-documented consequence: blue light emitted by smartphone screens suppresses melatonin production, delaying sleep onset by two to three hours and shifting the user's circadian rhythm [1]. Physical and psychological health effects include musculoskeletal discomfort, elevated rates of anxiety, and depression among heavy users [1].

2.3 Existing Intervention Strategies and Their Limitations

Software-based focus applications implement time-bounded blocking of selected applications. Their key vulnerability is the ease with which they can be overridden, as all controls reside on the device being regulated. This design contradiction is reflected in the high abandonment rates noted above [6,7], and is consistent with the broader pattern of disengagement from software-only behavioral interventions. Hardware-augmented solutions, such as Brick and Bloom, require physical interaction with an external object to activate focus mode, substantially raising the cost of impulsive override. Empirical evidence indicates that these products are more successful at sustaining focus sessions. However, they typically lack reward systems or gamification features that promote long-term habit formation [8,9].

2.4 Gamification and Habit Formation

Research shows that adding point systems, streaks, or challenge mechanics to productivity applications can increase user retention by up to 30 percent [8]. Case studies in the behavioral-app domain demonstrate that enhanced reward features can raise 30-day retention from 3.7 percent to 12.4 percent [9]. Personalized notifications and real-time rewards have been shown to measurably increase daily session frequency and habitual engagement [10,11]. The virtual pet mechanic is grounded in the psychological principle of nurturing investment: users who form an emotional attachment to a virtual entity become motivated to protect and grow it through sustained behavioral change. The micro-habit inducer component is similarly grounded in habit-stacking theory: the pairing of a new desired behavior with an existing behavioral trigger to accelerate its automatization [1,4]. Taken together, the literature strongly supports a design strategy that combines physical-activation bypass prevention with a multi-layered gamified reward system. An important caveat must be acknowledged, however: poorly calibrated gamification can itself reproduce compulsive engagement, shifting motivation from genuine habit formation to anxious score-protection [8,9]. Lock-In addresses this through the principle of sustainable engagement — rewards are tied exclusively to focus outcomes rather than app-interaction frequency, and the companion operates as a background motivator rather than a persistent demand for the user’s attention.

3. Design Requirements

This section presents the design requirements for the Lock-In application, derived through three complementary methodologies: a user needs survey, an analysis of existing solutions, and the formulation of functional and non-functional system specifications.

3.1 User Research and Findings

To validate the problem space and inform the feature set of Lock-In, the project team conducted a structured online survey targeting young adults and students interested in productivity. The survey was administered in both English and Thai to maximize accessibility. A total of 87 responses were collected across 21 questions organized over four thematic areas: demographics, smartphone usage patterns, interest in proposed features, and expected outcomes. Questions were developed in consultation with a faculty psychiatrist to ensure clinical relevance.

The majority of respondents (approximately 70 percent) were aged 18 to 24 and identified as students. Reported daily phone usage averaged between seven and nine hours, of which respondents estimated 55 percent as unproductive. Primary sources of distraction were identified as social media, messaging applications, and video streaming services. Figure 2 summarizes respondents' reported daily smartphone usage habits across key dimensions.

Figure 2. Survey Results — Daily Smartphone Usage Habits (n=87)

A significant proportion of respondents reported either having never used a focus application or having used one and discontinued, citing ease of bypass and lack of engaging features as the primary reasons for abandonment. Figure 3 summarizes prior focus-app adoption and satisfaction across the respondent pool.

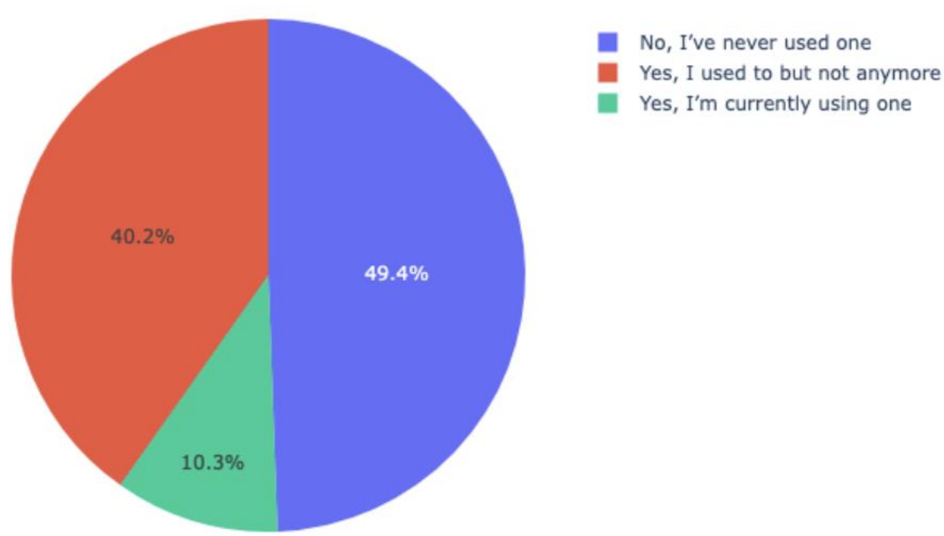

Figure 3. Survey Results — Prior Focus App Adoption (n=87)

Respondents expressed strong interest in personalized app-block durations, physical activation devices, virtual pet companions, and social leaderboard features, with a clear preference for positive reinforcement over punitive or restrictive approaches.

3.2 Analysis of Existing Solutions

A systematic review of the existing focus-application landscape was conducted across five representative products: Forest, Freedom, Focusmate, Brick, and Bloom. Each product occupies a distinct niche within the productivity and digital-wellbeing space, but each also exhibits specific limitations in either bypass resistance, motivational sustainability, or breadth of feature coverage. The following subsections provide a detailed analysis of each application. A consolidated feature comparison is presented at the end of this section.

3.2.1 Forest — Seekrtech (2015)

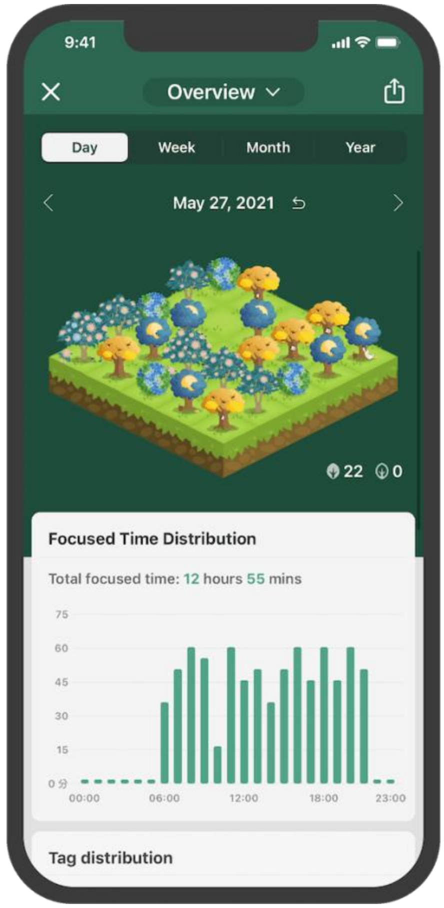

Figure 4. Forest application preview

Forest is a Pomodoro-style focus application available on iOS, Android, and as a browser extension. Its central mechanic is a virtual tree that begins growing when the user starts a focus session and dies if the user navigates away from the application before the session ends. Over time, users accumulate a virtual forest that visually represents their total focus history. Coins earned through completed sessions can be used to plant real trees through a partnership with the charity Trees for the Future, providing a tangible prosocial incentive layer.

Forest's core strengths are its approachable gamification metaphor and its cross-platform availability. The application also supports a friends feature through which users can grow shared forests, introducing a lightweight social accountability element. However, its app-blocking capability is limited: on Android it operates via the Accessibility Service API and can be disabled by the user at any time with minimal friction; on iOS, background app restrictions prevent any enforcement of blocking, rendering the system entirely honour-based. The gamification is constrained to the single tree-growth metaphor, which research suggests loses novelty within the first two to four weeks of use[8]. There is no micro-habit mechanism, no physical activation barrier, and no per-app statistics or distraction reporting.

3.2.2 Freedom — Freedom.to (2015)

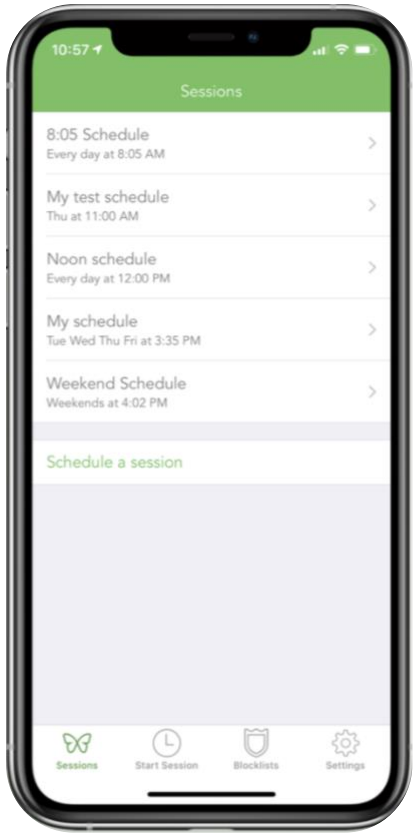

Figure 5. Freedom application preview

Freedom is a subscription-based distraction-blocking platform that operates across iOS, Android, macOS, Windows, and Chrome. Users define blocklists of websites and applications and schedule recurring or manual blocking sessions. A "Locked Mode" option prevents the user from disabling an active session before it expires, providing stronger enforcement than most software-only alternatives. Freedom's cross-device synchronisation means that a single block session simultaneously restricts access on all paired devices, which is a meaningful advantage for users who context-switch between desktop and mobile.

Despite its technical robustness, Freedom is a purely restrictive tool. It provides no reward, gamification, or positive-reinforcement mechanism of any kind: the user experience is one of removal rather than motivation. This design philosophy is intentional but creates a well-documented motivational vulnerability: without an intrinsic reward for completing a session, users who are sufficiently frustrated will disable the app, accept the financial cost of the subscription, or simply use a second device.[9] Freedom also offers no social or accountability features, no statistics beyond session logs, and no physical activation layer. Its pricing (approximately USD 40 per year) places it among the more expensive options in the category.

3.2.3 Focusmate — Focusmate Inc. (2017)

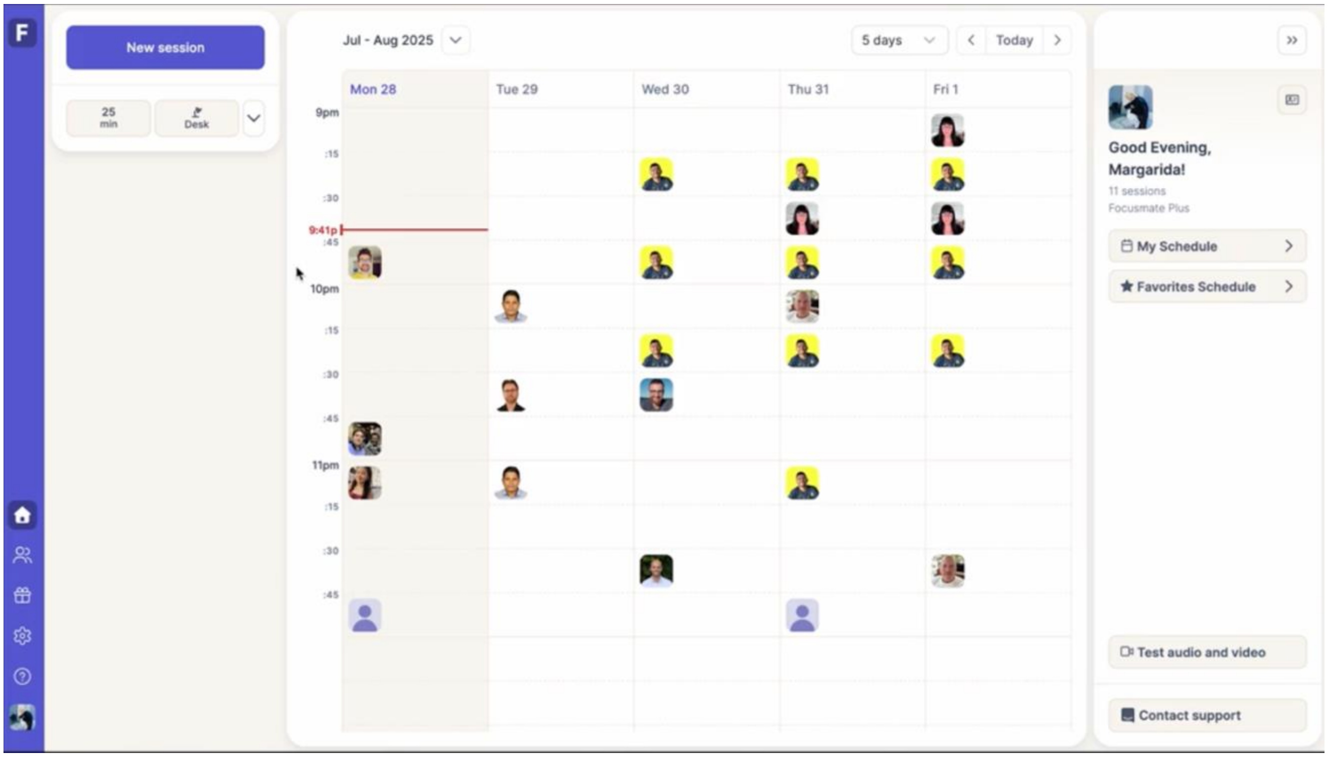

Figure 6: Focusmate website preview

Focusmate is a virtual co-working platform that pairs users with a live accountability partner for structured work sessions of 25, 50, or 75 minutes via video call. At the start of each session, both participants state their goals; at the end, they briefly report on their progress. The social contract created by the presence of another person has been shown to significantly reduce procrastination and increase task completion rates, a phenomenon known in behavioural science as body doubling.

Focusmate's primary limitation is its dependency on external human availability. Sessions must be booked in advance and are contingent on a partner showing up; time-zone constraints further reduce flexibility for users outside North American or European schedules. The platform provides no app-blocking capability — a user can freely browse social media during a Focusmate session — and the requirement for a live video feed can feel intrusive or stressful for users who prefer solitary working environments. The free tier is capped at three sessions per week, and the platform offers no gamification, no statistics dashboard, and no micro-habit mechanism.

3.2.4 Brick (2022)

Figure 7: Brick application and NFC device preview

Brick is a hardware-software hybrid focus system centred on a physical NFC card that must be physically tapped against the user's Android device to activate or deactivate focus mode. Once activated, a user-configured list of applications is blocked at the system level until the card is tapped again. The deliberate physical friction of requiring a tangible object to override the block substantially increases the cost of impulsive bypassing relative to purely software-based solutions: accessing a blocked app requires the user to physically retrieve the card, scan it, and consciously acknowledge the decision to end the session.

Brick's architecture directly addresses the bypass vulnerability that undermines software-only applications, making it the closest existing product to Lock-In in terms of physical activation philosophy. However, Brick is designed as a purely restrictive tool: there is no gamification layer, no companion or reward system, no social leaderboard, and no micro-habit mechanism. Its statistics are limited to basic session counts, and the card itself must be purchased separately at an additional cost of approximately USD 30–40. The absence of positive reinforcement means that Brick relies entirely on the user's pre-existing motivation, which prior research identifies as a primary cause of long-term app abandonment.[6]

3.2.5 Bloom — Bloom (2023)

Figure 8: Bloom application and NFC device preview

Bloom is a focus and digital-wellness application that combines a physical NFC- or Bluetooth-enabled device with mobile app blocking. Similar to Brick, the physical device must be engaged to activate a focus session, and the paired mobile application enforces a configurable app blocklist during active sessions. Bloom differentiates itself from Brick by incorporating a basic break-management system: the application surfaces scheduled short-break reminders and allows the user to configure break durations alongside session lengths, providing a more structured session rhythm aligned with evidence-based work-rest cycles.

Despite these additions, Bloom's feature coverage remains limited. Its gamification layer is minimal — there is no companion pet, coin economy, or progression system that creates ongoing intrinsic motivation beyond the session itself. The social dimension is absent: there is no leaderboard, no friends system, and no mechanism for peer accountability. Statistics are limited to basic session-length logs without per-application distraction breakdowns or trend visualisation. Like Brick, the hardware component represents an additional purchase, and the app's wellness-oriented design language, while approachable, does not provide the sustained motivational scaffolding that the target user population — adolescents and young adults — has been shown to respond to in gamified productivity contexts.[10,11]

In summary, the five analyzed products each address one or two dimensions of the focus-application problem space but none integrates physical bypass resistance, comprehensive gamification, social accountability, and behavioral micro-intervention within a single cohesive system. This gap defines the core design opportunity for Lock-In. The following table provides a consolidated feature-by-feature comparison across all five competitors.

Feature	Forest	Freedom	Focusmate	Brick	Bloom	Lock-In
Physical Activation (NFC / Hardware)	✗	✗	✗	✓	✓	✓
App Blocking	~	✓	✗	✓	✓	✓
Focus Timer	✓	✓	✓	✓	✓	✓
Gamification / Virtual Companion	✓	✗	✗	✗	✗	✓
Social Leaderboard & Friends	~	✗	✓	✗	✗	✓
Micro-habit Inducer	✗	✗	✗	✗	✗	✓
Break Management	✓	✗	✗	✗	~	✓
Statistics Dashboard	✓	~	~	~	~	✓

Legend: ✓ Full support ~ Partial support ✗ Not available

3.3 System Requirements

3.3.1 Functional Requirements

Based on the user research, competitive analysis, and project scope defined in the pre-project proposal, the following functional requirements were established:

Category	Requirement
App Blocking	The system shall block user access to applications on a configurable blacklist during an active focus session.
App Blocking	The system shall display a dedicated overlay screen when a blacklisted app is accessed, preventing navigation into the app.
Focus Timer	The system shall allow users to configure and start a focus session of a specified duration.
Focus Timer	The system shall display a real-time countdown timer during an active focus session.
Focus Timer	The system shall allow users to schedule automatic notification for recurring focus sessions.
NFC Activation	The system shall require a physical NFC tap to activate a focus session.
NFC Activation	The system shall support NFC device pairing with a user account.
NFC Activation	The system shall implement a fallback mechanism in the event that NFC detection fails.
Micro-habit Inducer	The system shall allow users to configure a preferred short physical action as an unlock condition during focus sessions.
Micro-habit Inducer	The system shall prompt the user to complete their configured micro-habit action before granting temporary phone access during focus time.
Virtual Pet	The system shall maintain a virtual companion whose growth state is determined by the user's cumulative focus time.
Virtual Pet	The system shall allow users to select from multiple companion types at onboarding.
Virtual Pet	The system shall allow users to purchase and apply decorations to the companion's room using focus-earned currency.
Statistics	The system shall record and display focus session history, including duration, date, and app-block events.
Statistics	The system shall compute and display aggregate metrics including total focus time, daily streak, and weekly trends.
Leaderboard	The system shall implement a social leaderboard ranking users by focus time within a defined period.
Leaderboard	The system shall allow users to add friends and view friend-specific leaderboard rankings.
Account	The system shall allow users to register, log in, and manage their account.
Schedule	The system shall allow users to create named focus schedules with configured duration, recurring days, and start time.

Table 1. Functional Requirements

3.3.2 Non-Functional Requirements

Category	Requirement
Performance	The application shall respond to user interactions within 2 seconds under normal network conditions.
Performance	App-blocking overlay activation shall occur within 500 milliseconds of the user navigating to a blacklisted app.
Reliability	The system shall implement an NFC fallback mechanism to maintain functionality when NFC hardware is unavailable.
Security	All user data transmitted between the client and server shall be encrypted in transit using HTTPS/TLS.
Security	User passwords shall be stored using a secure hashing algorithm. The system shall protect against SQL injection and XSS.
Usability	The application shall be operable by users with no prior experience of the app within a single session, without requiring external documentation.
Privacy	The application shall collect only data necessary to its core functionality. No user behavioral data shall be shared with third parties.
Portability	The application shall be compatible with Android devices running Android 10 (API level 29) and above.
Scalability	The backend architecture shall support horizontal scaling to accommodate user growth beyond initial deployment.
Ethics	All gamification and reinforcement mechanisms shall be designed to support positive reinforcement only. Punitive or manipulative behavioral design patterns shall be avoided. Reward mechanisms shall follow the principle of sustainable engagement: growth and rewards shall be tied to focus outcomes rather than application-interaction frequency, so that the system does not replace one form of compulsive phone use with another.

Table 2. Non-Functional Requirements

3.4 Architecture Decision

Two primary implementation architectures were evaluated: an entirely on-device model and a cloud-based model with a backend server.

The on-device model stores all user data locally within the application's memory, requiring no backend service. While simpler to implement, this approach introduces a high risk of data loss upon device replacement or application uninstallation and cannot support cross-device synchronization or social features such as the leaderboard.

The cloud-based model introduces a backend service and cloud database, enabling persistent and synchronized user data, multi-device NFC registration, and the full suite of social features. The added complexity is justified by the project's requirements for a leaderboard and robust data retention.

The cloud-based architecture was selected as the implementation model for Lock-In. The technology stack chosen for implementation is as follows: Flutter with Dart for the cross-platform Android frontend; ExpressJS or NestJS with TypeScript for the backend service (with Firebase as a fallback Backend-as-a-

Service option); PostgreSQL for structured relational data (with Cloud Firestore as an alternative for Firebase deployment); and AWS, Firebase, or Render for cloud hosting and deployment. Version control and CI/CD are managed through GitHub and GitHub Actions. UI and UX design prototyping is conducted in Figma.

3.5 Feature Prioritization

The development of the "Locked In" ecosystem follows a phased implementation strategy, prioritizing features based on a synthesis of user psychological needs and technical criticality.

Feature	Type	Priority	Description
Restrictive App Blocking	Core Foundation	High	Implementation of the native overlay and usage monitoring service; the primary behavioral "stick."
Authentication & Identity Management	Core Foundation	High	Secure onboarding and session persistence using JWT to establish user-system trust.
Basic Pet Growth Engine	Core Foundation	High	Foundational XP logic and state machine to provide an immediate feedback loop.
Onboarding & Personalization	Core Foundation	High	Mechanism for companion selection to establish the initial emotional bond with the user.
Advanced Mode Configuration	Behavioral Reinforcement	High	Granular control over app blacklists configurations for varying contexts.
NFC Hardware Integration	Behavioral Reinforcement	Medium	Integration of nfc_manager to require physical activation for focus sessions.
Focus Scheduling	Behavioral Reinforcement	Medium	Scheduled notification for user to start focus periods to enable proactive time management.
App Insights & Analytics	Behavioral Reinforcement	Medium	Insights into productivity trends to encourage habit-forming meta-cognitive awareness.
Social Connectivity & Leaderboards	Social & Aesthetic Expansion	Low	Implementation of social motivation through community accountability and competition.
Virtual Environment Customization	Social & Aesthetic Expansion	Low	The "Room & Decorations" system providing a meaningful use for coins and representation of success.
Interactive Pet Mechanics	Social & Aesthetic Expansion	Low	Specialized animations (eating, dancing, petting) to deepen the user-pet interactions.

Table 3. Feature Priority

4. Design Description

The Lock-In System is designed as a modular, mobile-centric application that integrates NFC-based device control, focus management, gamification elements, and social features. The architecture follows a layered client–server model with clear separation between presentation, business logic, and data persistence. The system emphasizes scalability, maintainability, and CI/CD automation to ensure consistent delivery and quality control.

4.1 Overview

The overall system architecture (see Figure 9) consists of three primary layers:

- Frontend (Mobile Application)
- Backend (RESTful API Server)
- Database & Storage Layer

The mobile application communicates with the backend through secure HTTP requests. Authentication is handled via JWT, and all incoming requests pass through validation and guard mechanisms before reaching business logic components. Additionally, app usage data and Android-specific functions are accessed through the Android API.

The backend follows a modular structure (Controller–Service pattern) and dependency injection pattern, enabling separation of concerns and easier scalability of features.

PostgreSQL serves as the primary relational database for structured and persistent data, with Redis integrated as an in-memory data store.

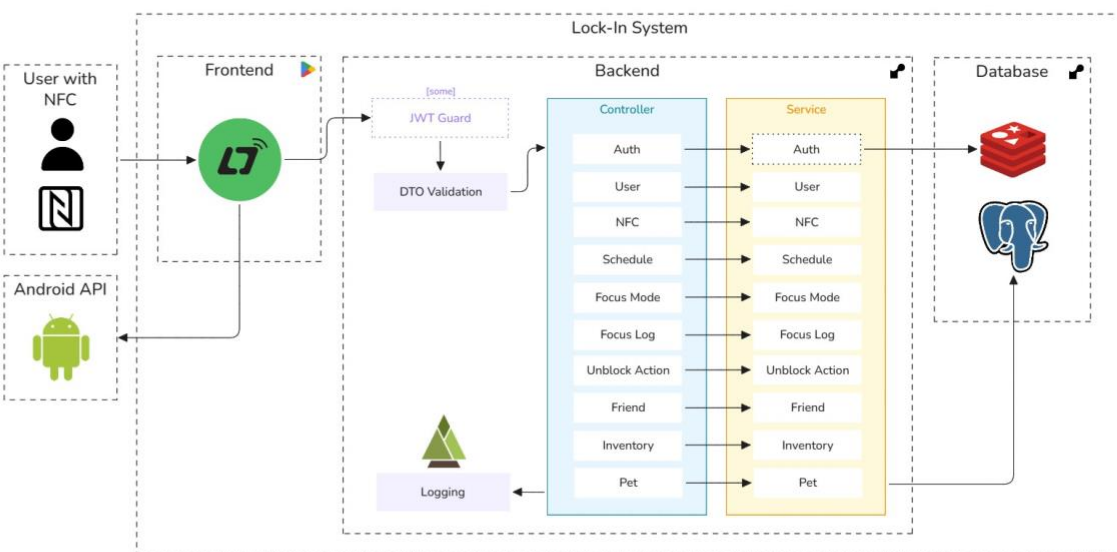

Figure 9. System Architecture Diagram

4.2 Detailed Description

4.2.1 Software Engineering Diagrams

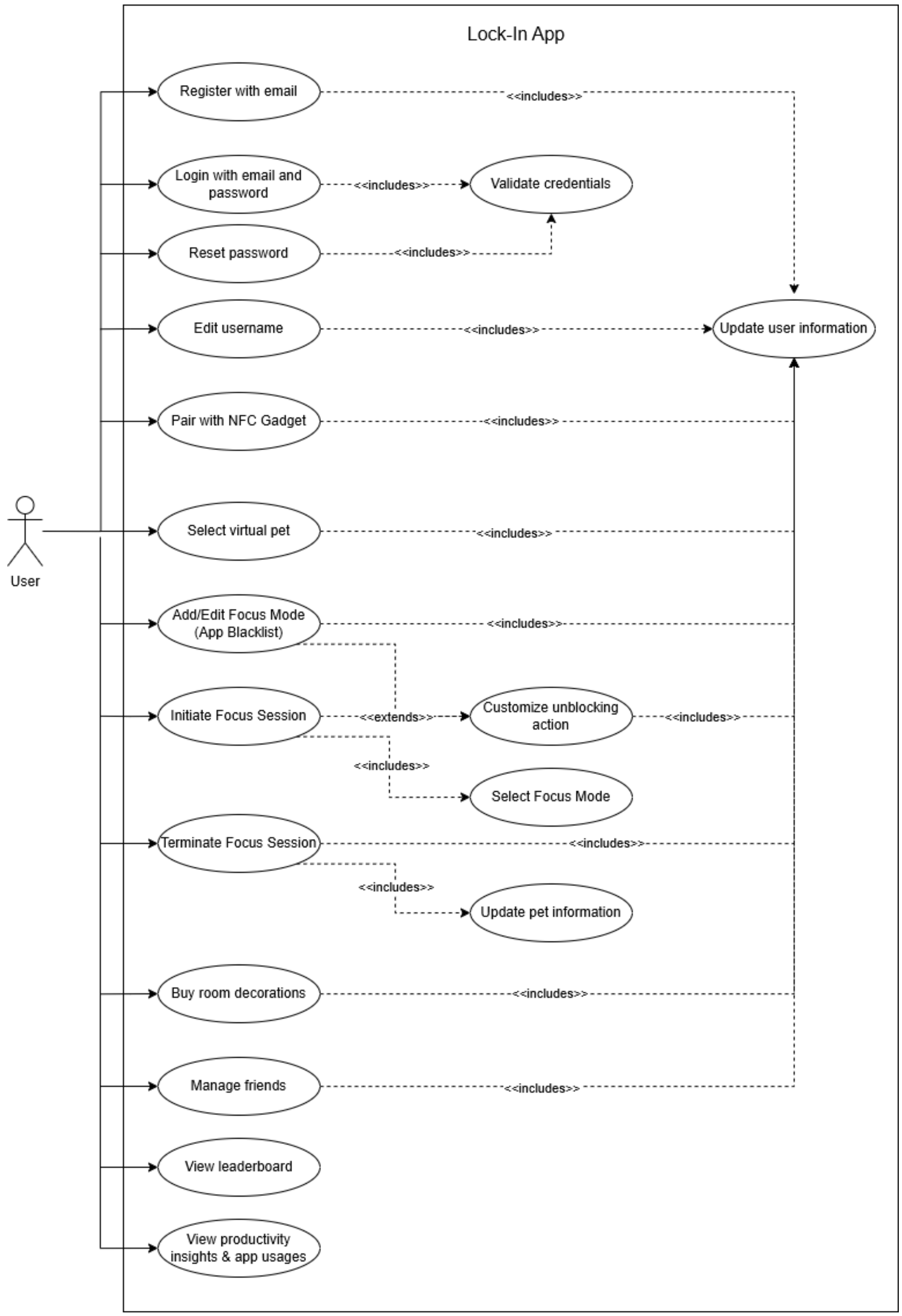

Figure 10. Use Case Diagram

4.2.2 Database Diagram

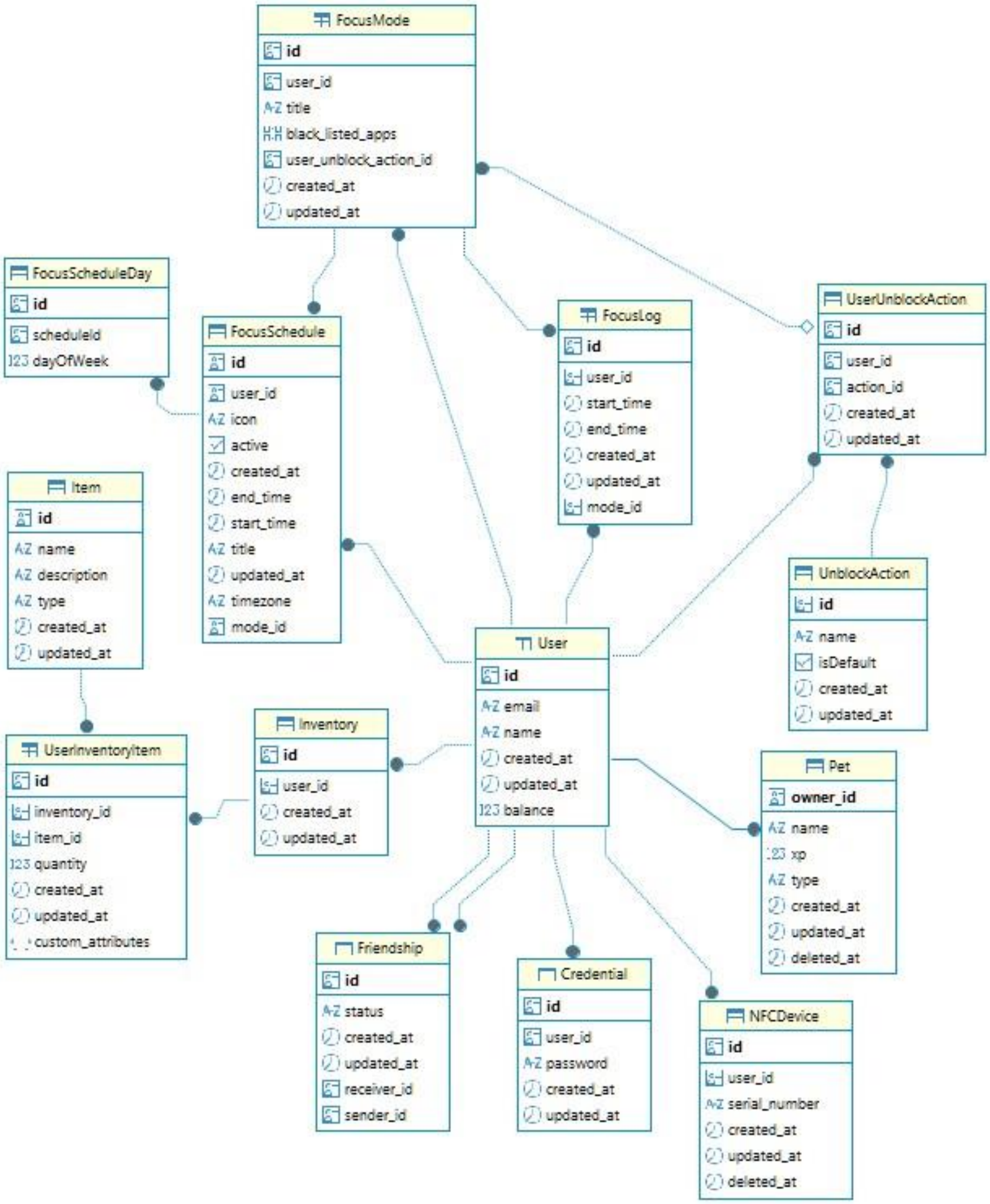

Figure 11. Database Entity Relationship Diagram

4.2.3 Design, Prototype, and Project Management

The Lock-In System follows an iterative and collaborative development approach. UI/UX design and interactive prototyping are conducted using Figma, allowing the team to validate user flows and interface consistency before implementation. This helps reduce redesign effort during development and ensures alignment between design and functionality.

System architecture diagrams, schemas, and workflow illustrations are created using Excalidraw, providing clear visualization of technical structures and data relationships. These diagrams support better communication and early validation of architectural decisions.

Source code for both frontend and backend is managed through GitHub, which supports version control, branching strategies, pull requests, and CI integration. Automated checks ensure code quality before merging into the main branch.

Project planning, documentation, and task tracking are handled using Notion, serving as a centralized workspace for requirements, sprint planning, and progress monitoring. Together, these tools enable structured development, clear documentation, and efficient team collaboration.

4.2.4 Frontend

The frontend of the Lock-In System is developed using Flutter, a cross-platform framework chosen for its ability to deliver high-performance, visually rich user interfaces from a single codebase. The application's architecture is built around a declarative UI approach, ensuring a responsive and consistent user experience across different devices.

State management is orchestrated via Riverpod 3.0, which provides a type-safe and testable foundation for managing global application state and asynchronous data flows. Navigation is handled by GoRouter 15.0, facilitating a modular routing system that supports deep linking and complex authentication-based redirection logic. To implement the virtual pet gamification features, the project integrates the Flame 1.25 engine, allowing for smooth sprite animations and interactive game elements to be embedded directly within the Flutter widget tree.

Networking is managed through the Dio client, utilizing interceptors for automated token management and error handling. For local data persistence, the system employs SharedPreferences for lightweight configuration and Drift for robust, structured relational storage. A key technical feature is the integration of native Android capabilities through Method Channels, which enables the application to perform system-level tasks such as app blocking. Furthermore, the frontend utilizes NFC technology for physical focus session initiation, leveraging the NDEF format to verify user presence at designated focus stations.

4.2.5 Backend

The backend system is developed using NestJS with TypeScript, selected for its modular structure and strong architectural conventions. NestJS follows a layered design pattern based on Controllers and Services, ensuring clear separation between request handling and business logic.

A key architectural characteristic of the backend is its dependency injection (DI) pattern, which is natively supported by NestJS. Through DI, services and components are injected into controllers and other modules via constructors, promoting loose coupling and improved testability. This approach enhances maintainability, simplifies unit testing through mock injection, and enables scalable module expansion without tightly binding components together.

Authentication is implemented using JWT-based guards to secure API endpoints. Incoming requests are validated using Data Transfer Objects (DTOs), ensuring input type and value safety before processing. Logging, via Pino, mechanisms are also integrated to support monitoring and debugging.

The backend is organized into domain-specific modules, including Auth, User, NFC, Schedule, Focus Mode, Focus Log, Unblock Action, Friend, Inventory, and Pet. Each module encapsulates its own controller and service layer, promoting modularity and reducing interdependencies.

For data persistence, the backend connects to PostgreSQL as the primary relational database and Redis as an in-memory cache to improve performance for frequently accessed or time-sensitive data. The application is containerized using Docker to ensure consistent environments across development and production, supporting scalable and reliable deployment.

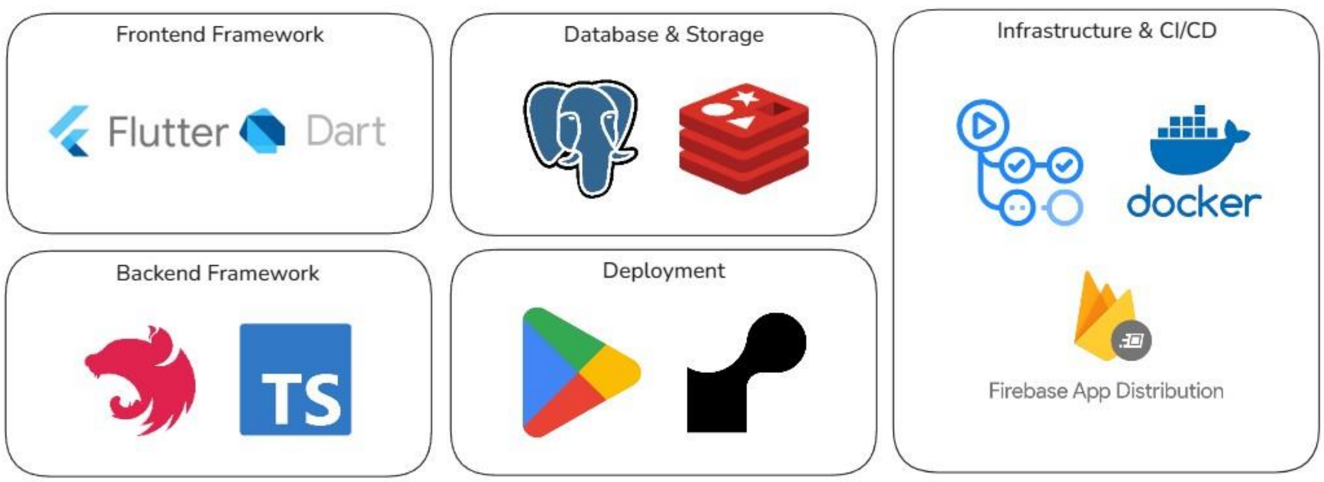

Figure 12. Technology Stack

4.2.6 AI Tools Utilization

The development of the "Locked In" ecosystem was significantly optimized through the strategic integration of advanced AI coding assistants, primarily **Claude Code** and **GitHub Copilot** (Codex). Rather than treating these tools as simple autocomplete utilities, they were orchestrated within a structured, human-in-the-loop framework to handle a broad spectrum of engineering tasks, ranging from initial requirement analysis to final documentation.

The project employed a tiered approach to AI utilization, selecting tools based on task complexity and computational efficiency:

- Claude Code: Serving as the primary autonomous agent, Claude Code managed high-level development workflows. This included task planning, implementation, and quality assurance. A pivotal aspect of this integration was the use of the Model Context Protocol (MCP), specifically the Figma MCP. This allowed the assistant to directly query design files to extract precise node hierarchies, spacing tokens, and typography. Consequently, the assistant could generate high-fidelity Flutter components that strictly adhered to the design system, ensuring a "pixel-perfect" UI across onboarding and focus screens.

- GitHub Copilot: Utilizing the Codex model, Copilot was deployed for granular, high-frequency tasks where token efficiency and rapid context handling were paramount. It was particularly effective for basic debugging, explaining legacy business logic, and generating documentation for minor features where the full agency of Claude Code was unnecessary.

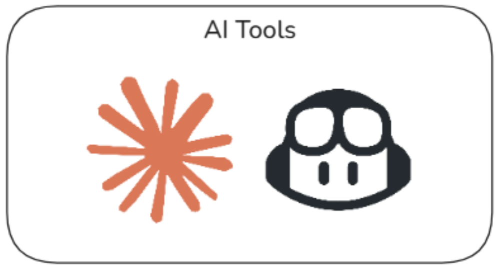

Figure 13. AI Tools

To maintain high technical standards, the assistants were directed through task-specific instructions, functioning as specialized behavioral agents across four key domains:

1. **Code Synthesis and Boilerplate:** Generating foundational code for Riverpod providers, Freezed models, and NestJS modules, thereby reducing manual entry for repetitive architectural patterns.

2. **Automated Test Generation:** Constructing comprehensive test suites, including unit tests for core business logic and widget tests for the UI, to facilitate early regression detection and ensure high code coverage.

3. **Surgical Code Reviews:** Performing critical evaluations of existing code to identify edge cases, suggest idiomatic improvements, and verify the consistent application of project-wide architectural standards.

4. **Technical Documentation:** Drafting API references and implementation guides to ensure structural consistency and technical accuracy throughout the project lifecycle.

Despite the high degree of automation, all AI-generated contributions remained under strict human supervision. Developers functioned as orchestrators, validating architectural decisions, verifying the logic of complex algorithms, and performing manual adjustments to ensure performance and security. This collaborative model maximized development velocity while ensuring the final implementation met the rigorous technical integrity required for the ecosystem.

4.2.7 Software Testing & Deployment

The backend system adopts an automated CI/CD pipeline (see Figure 15) to ensure code quality, reliability, and consistent deployment. The pipeline is triggered whenever a pull request is created or updated against the main branch. This enforces automated validation before any code can be merged into production.

The CI process begins with dependency installation and environment setup, followed by a series of automated checks. These include linting to enforce coding standards, unit testing to verify business logic correctness, and application build validation to ensure the API compiles successfully. Database build verification is also performed to confirm schema compatibility and prevent migration-related issues. Only when all stages pass successfully is the pull request eligible for merging.

Once merged, the CD process automatically builds a Docker image of the backend application and proceeds with deployment to the production server. Containerization ensures environmental consistency across development and production, minimizing deployment discrepancies.

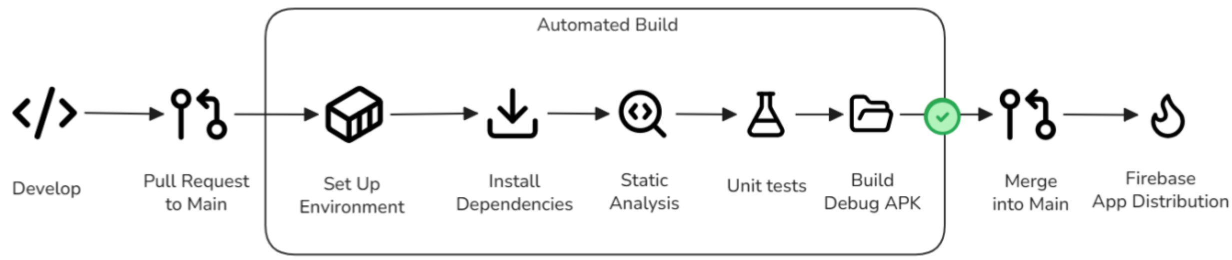

Figure 14. Frontend CI/CD Pipeline

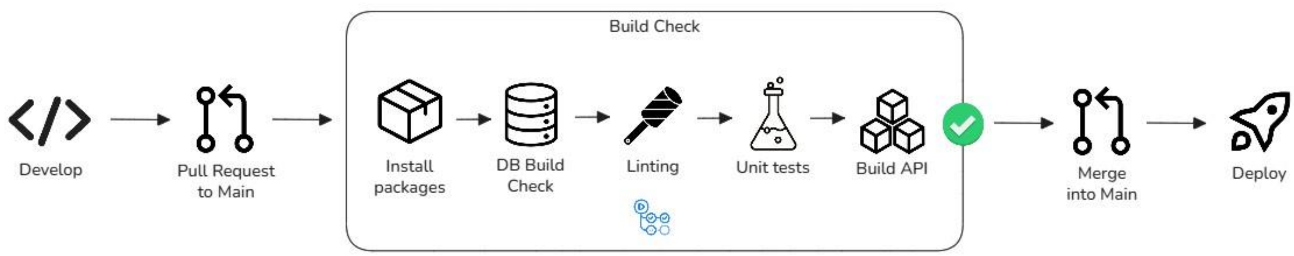

Figure 15. Backend CI/CD Pipeline

4.3 Use

The system design of Lock-In is centered on a minimalist, high-contrast interface intended to reduce cognitive load and enhance readability for users seeking to manage screen time. The platform integrates hardware-based interaction with gamification to foster a structured focus environment that facilitates productivity for various user groups. The following subsections detail the core functional flows and the UI/UX principles applied to each module, demonstrating a clear visual hierarchy and progressive interaction design.

4.3.1 Onboarding & Account Creation / Login

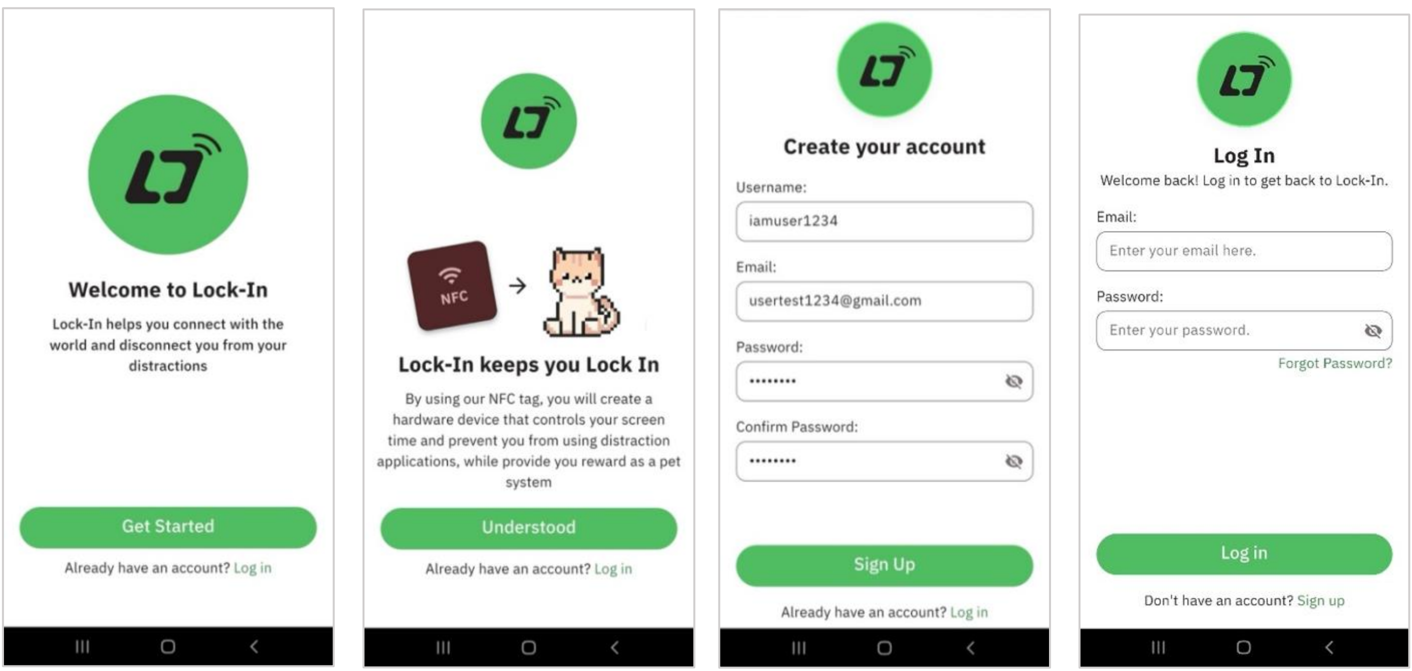

Figure 16. Onboarding & Account Creation / Login Flow.
(a) Landing; (b) NFC Notice; (c) Sign Up; (d) Log In.

The user journey begins with an interface-driven interaction designed to guide users smoothly toward core functionality. The landing screen (see Figure 16a) presents a clean layout featuring the application logo and a brief description of the system's purpose. The interface includes primary call-to-action buttons such as "Get Started" for new users and "Already have an account? Log in" for returning users. A dedicated orientation screen (see Figure 16b) introduces the NFC tag as a hardware device that controls screen time while providing rewards through a virtual pet system. The account creation interface (see Figure 16c) includes clearly labeled fields for username, email, and password confirmation to ensure ease of use. Correspondingly, the login page (see Figure 16d) is designed with a minimalistic interface that requires only essential information to reduce friction and onboarding time.

4.3.2 Companion Selection

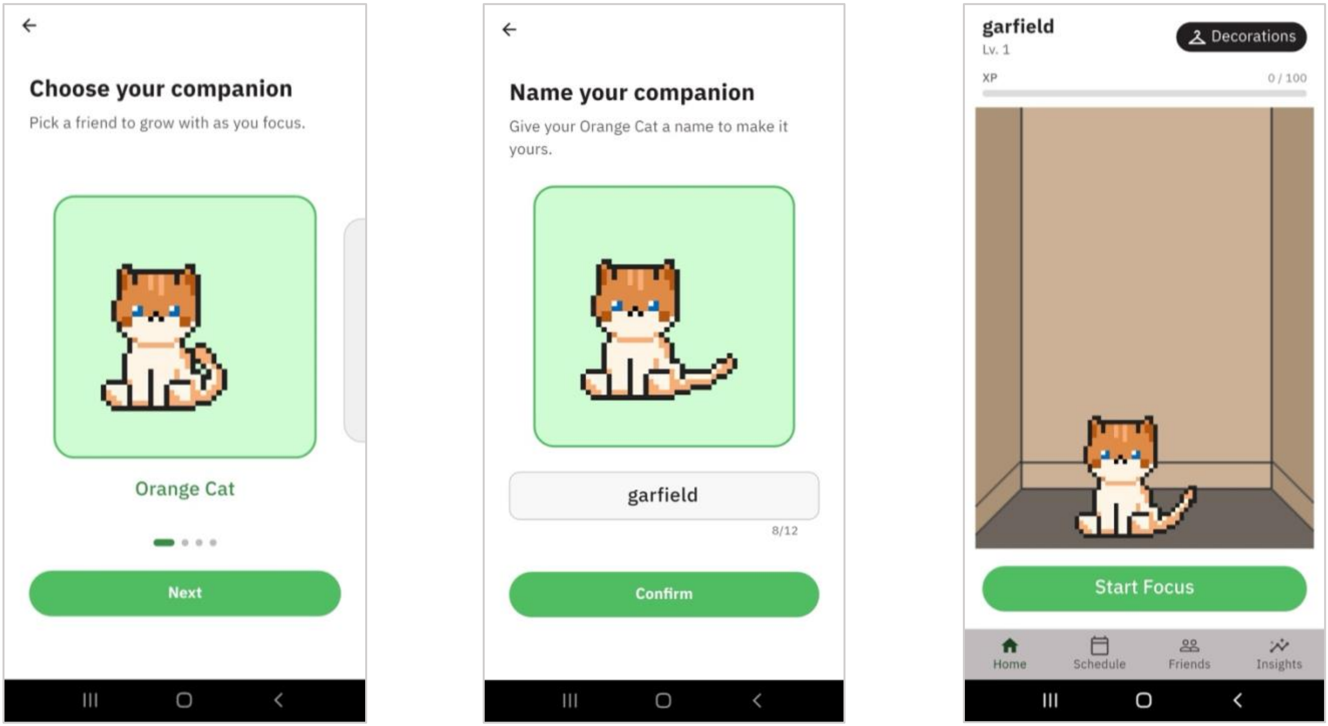

Figure 17. Companion Selection Flow.
(a) Companion Type Selection; (b) Companion Naming; (c) Home Page.

Following authentication, the system transitions to a personalization phase intended to establish an emotional connection with the user. Users are presented with a selection of virtual companions (see Figure 17a) displayed in a pixel-art style that creates a distinct personality. The interface (see Figure 17b) allows users to assign a custom name to their chosen companion within a text input field that features a character count indicator for clarity. Upon confirming the name, the system redirects the user to the Home Page (see Figure 17c), where the companion is positioned prominently in a room-like environment alongside an experience (XP) progress bar and the "Start Focus" call-to-action button. Persistent navigation at the bottom of the screen provides seamless access to the Home, Schedule, Friends, and Insights modules.

4.3.3 NFC Connection

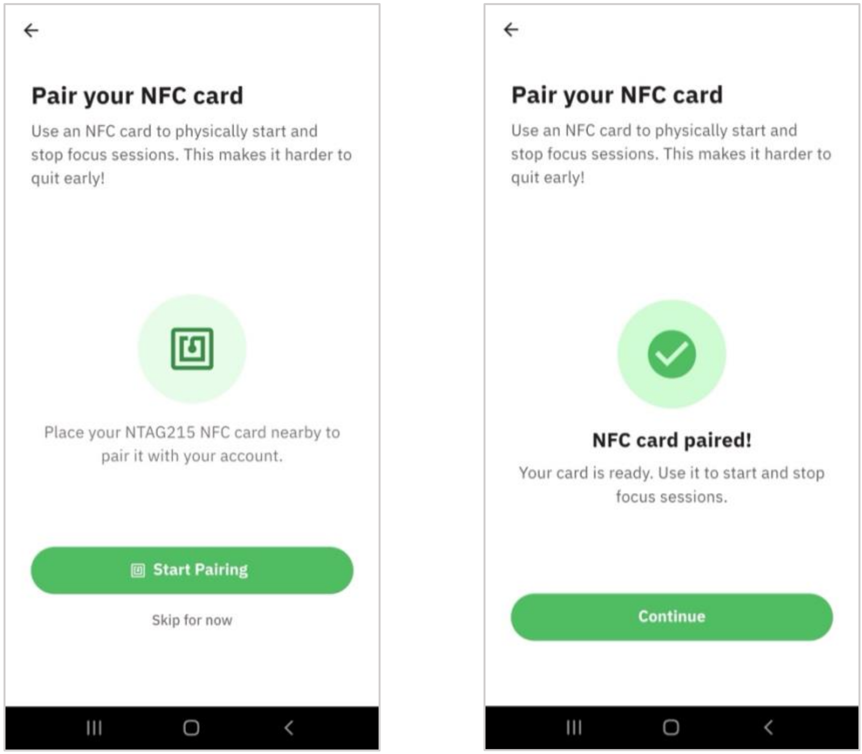

Figure 18. NFC Connection Screens. (a) Start Pairing; (b) Pairing Success.

To ensure hardware-based control and strengthen user commitment, the system incorporates an NFC pairing process. Users are provided with clear instructions to use an NFC card to physically start and stop focus sessions, which makes it intentionally harder to quit tasks early. The interface (see Figure 18a) prompts the user to place an NTAG215 NFC card nearby to pair it with their account. Successful interaction (see Figure 18b) is visually confirmed through a checkmark icon and the message "NFC card paired!", providing the user with immediate feedback that the hardware is ready for operation. This physical interaction layer is designed to enhance behavioral accountability.

4.3.4 Create / Edit a Focus Schedule

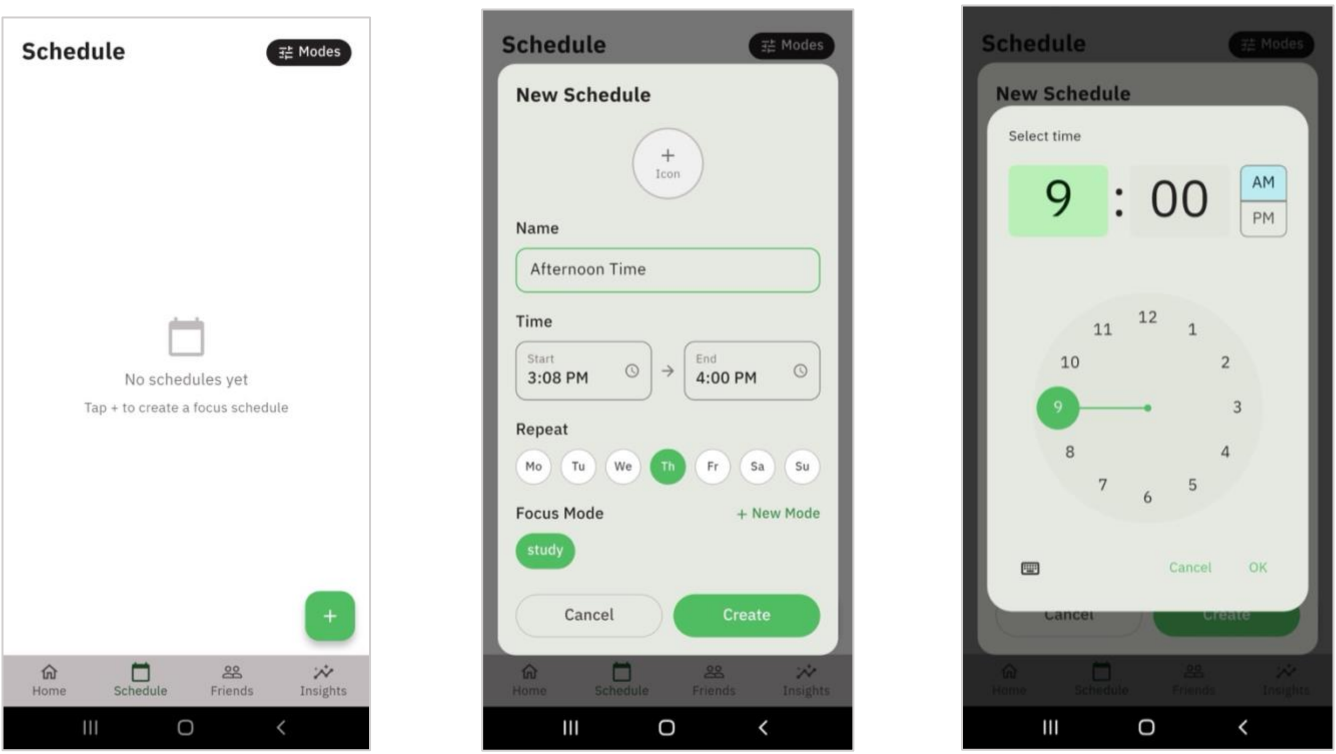

Figure 19. Schedule Creation Flow. (a) Empty Schedule; (b) New Schedule Modal; (c) Time Picker.

The scheduling module enables users to automate their productivity routines, by utilizing a scheduled notification at desired time, through a structured configuration interface. Initially, the main interface (see Figure 19a) displays a placeholder message when no schedules exist, encouraging the user to tap a floating action button to create a new entry. Users can define a "New Schedule" by specifying a name and setting a repetition pattern across various days of the week (see Figure 19b.) An interactive circular clock time picker (see Figure 19c) is provided to allow for intuitive selection of start and end times. Each schedule is linked to a specific Focus Mode, ensuring that predefined restrictions are automatically applied at the designated time.

4.3.5 Create New / Edit Mode

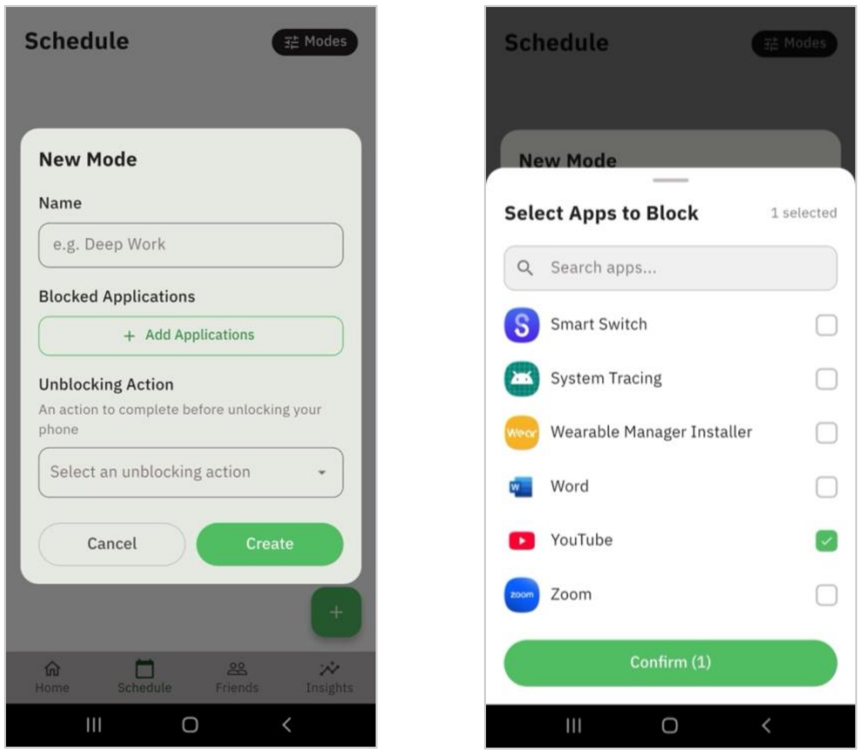

Figure 20. Mode Creation Flow. (a) New Mode Modal; (b) Blocked App Selection Sheet.

Customization of the focus environment is managed through a dedicated mode creation interface that utilizes modal windows for task efficiency (see Figure 20a.) Users begin by naming their mode and then proceed to select specific "Blocked Applications" from a comprehensive list. The selection sheet (see Figure 20b) includes checkboxes for various installed apps and provides a confirmation button that reflects the number of selected items to be restricted. Furthermore, the system introduces a behavioral commitment through an "Unblocking Action," where the user must select or define a specific task, like "drink water," that must be performed before the phone can be unlocked at the end of a session.

4.3.6 Start a Focus Session

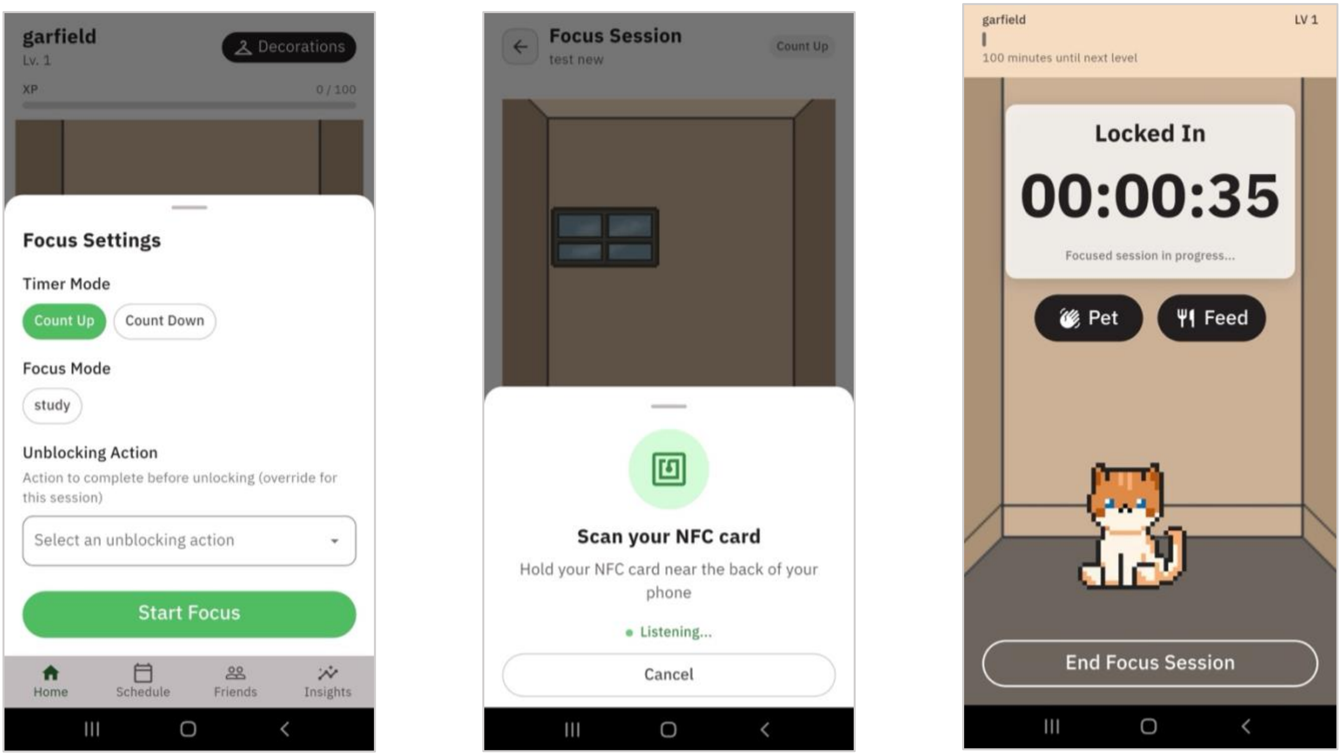

Figure 21. Start Focus Session Flow.
(a) Focus Settings Sheet; (b) NFC Activation Sheet; (c) Focus Screen.

In the Focus Settings sheet (see Figure 21a), users select their preferred timer mode, either "Count Up" or "Count Down", and their desired focus mode. The system then prompts physical verification (see Figure 21b), requiring the user to hold their NFC card near the back of the phone while a "Listening..." indicator provides real-time status. Once active, the "Locked In" screen (see Figure 21c) displays a large timer showing progress. During the session, users can engage in micro-interactions with their pet, such as "Pet" or "Feed," which reinforces the gamified bond without providing a significant distraction from their primary focus.

4.3.7 Taking Short Break

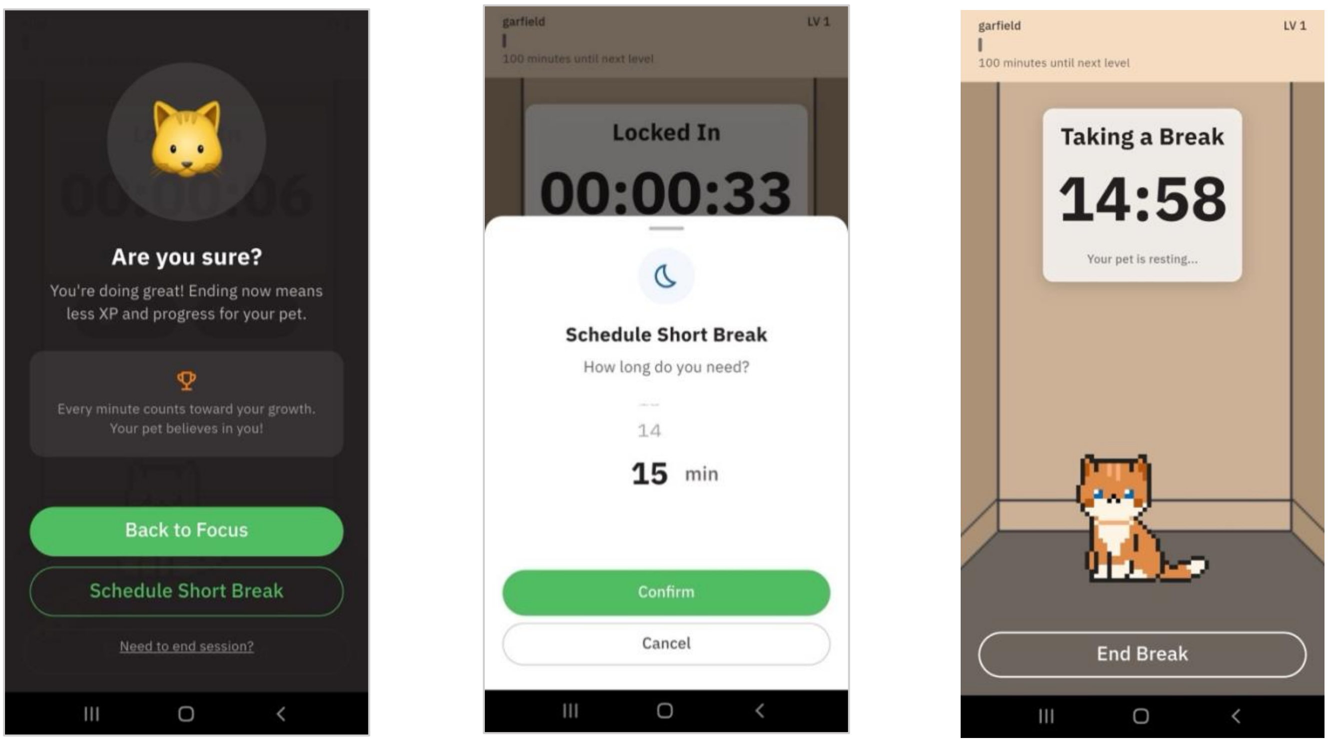

Figure 22. Short Break Flow.
(a) *(a) End Session Overlay; (b) Schedule Short Break Modal; (c) Break Screen.*

To prevent burnout, the system supports structured rest periods within a controlled workflow. Attempting to exit an active session triggers a confirmation overlay that warns the user that ending prematurely will result in less XP and slower progress for their virtual pet (see Figure 22a). If a user chooses to "Schedule Short Break," they are presented with a modal interface to select a break duration (see Figure 22b). During the break, the interface (see Figure 22c) shifts to a "Taking a Break" state, displaying a countdown timer and a visual representation of the pet resting. This mechanism balances enforced productivity with the necessity of scheduled downtime.

4.3.8 Stop/Complete a Focus Session

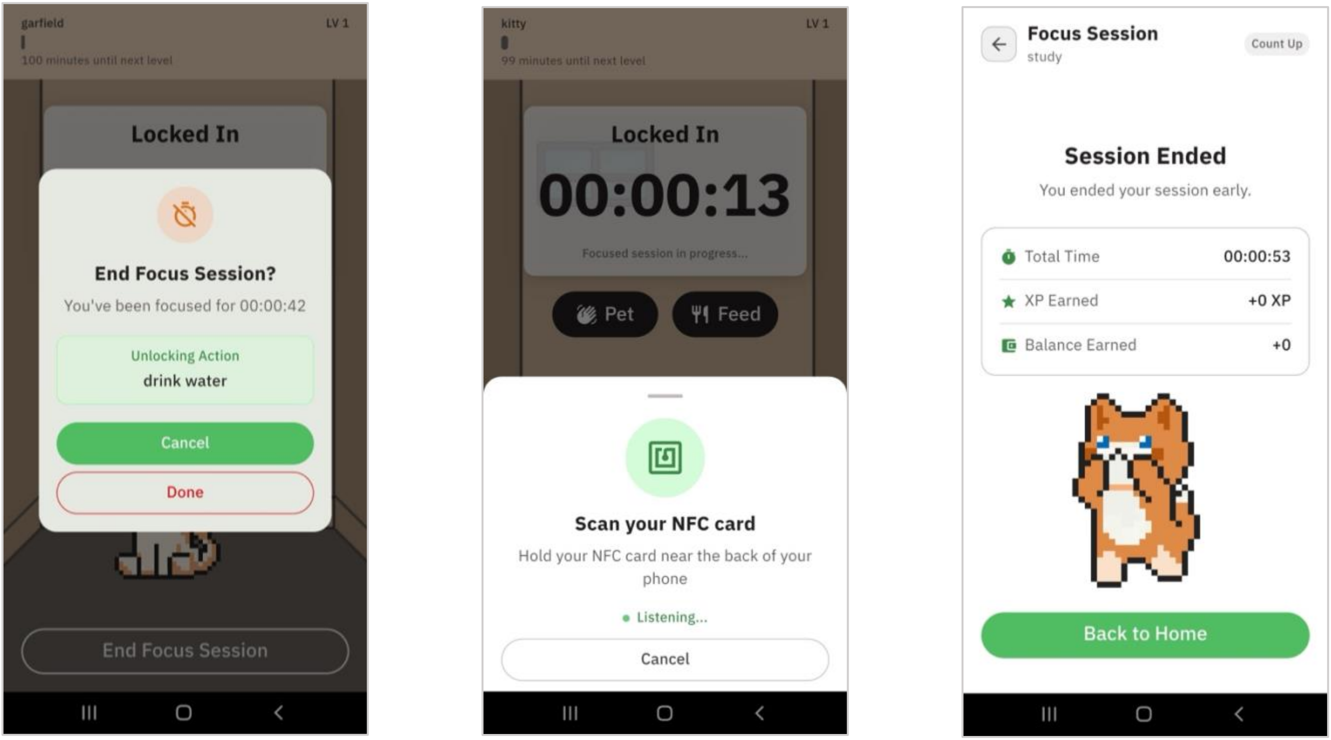

Figure 23. Stop Focus Session Flow.
(a) Confirmation Modal; (b) NFC Activation Sheet; (c) Session Ended Screen.

Concluding a focus session requires intentional effort and provides users with analytical feedback on their performance. When a session is ended, the user must often perform the predefined Unblocking Action (see Figure 23a) and use the paired NFC card to end the session (see Figure 23b). The "Session Ended" screen (see Figure 23c) provides a detailed summary, including metrics such as Total Time focused, XP Earned, and Balance Earned. If a session ends early, the system provides immediate negative reinforcement by showing a small amount of XP, as calculated from focus session duration, thereby highlighting the loss of progress and encouraging the completion of future sessions.

4.3.9 Add Friends and See Leaderboard

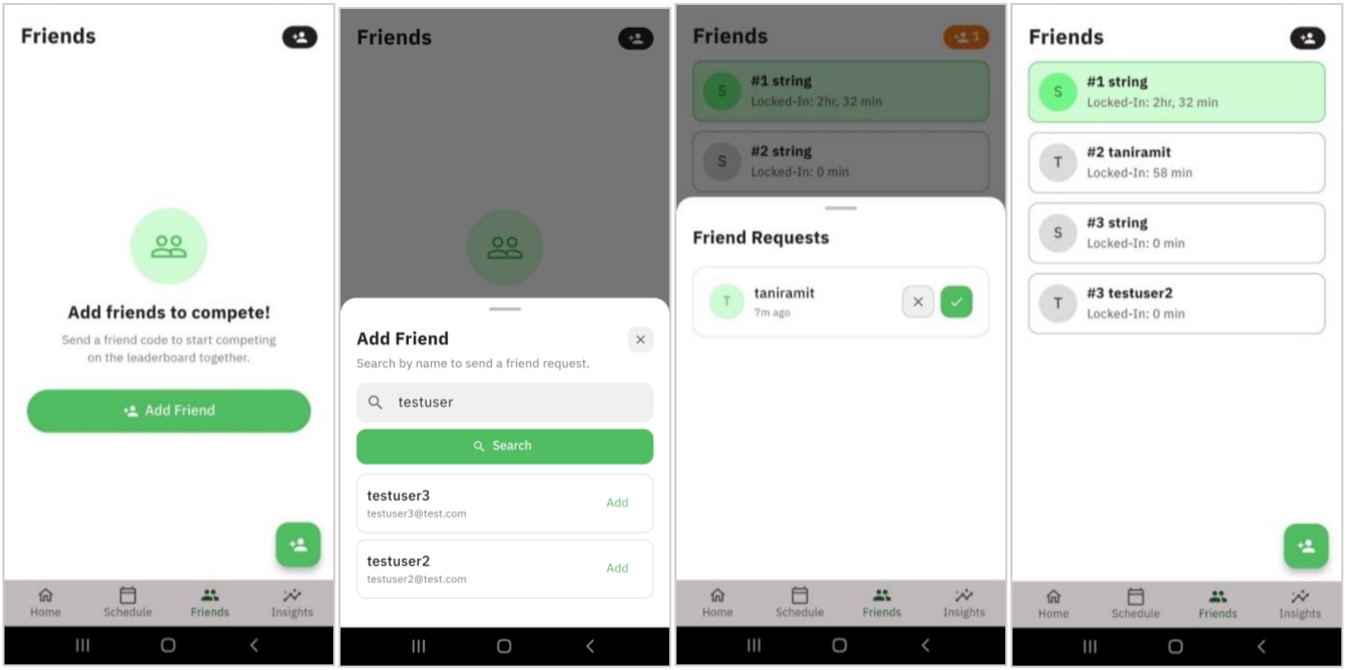

Figure 24. Friend & Leaderboard Screens.
(a) *(a) Empty Friend Screen; (b) Friend Search Sheet; (c) Friend Requests Sheet; (d) Leaderboard.*

The social module of Lock-In encourages consistent engagement through a competitive community framework (see Figure 24a). Users can search for other members by name using a search sheet and send friend requests to build their network (see Figure 24b). Incoming requests are managed through a dedicated list where they can be accepted or declined (see Figure 24c). The system features a Leaderboard that ranks connected friends based on their total "Locked-In" time (see Figure 24d). Each entry on the leaderboard displays the user's status and focus time, fostering a sense of accountability and social motivation to maintain high productivity levels.

4.3.10 Decorate Your Room

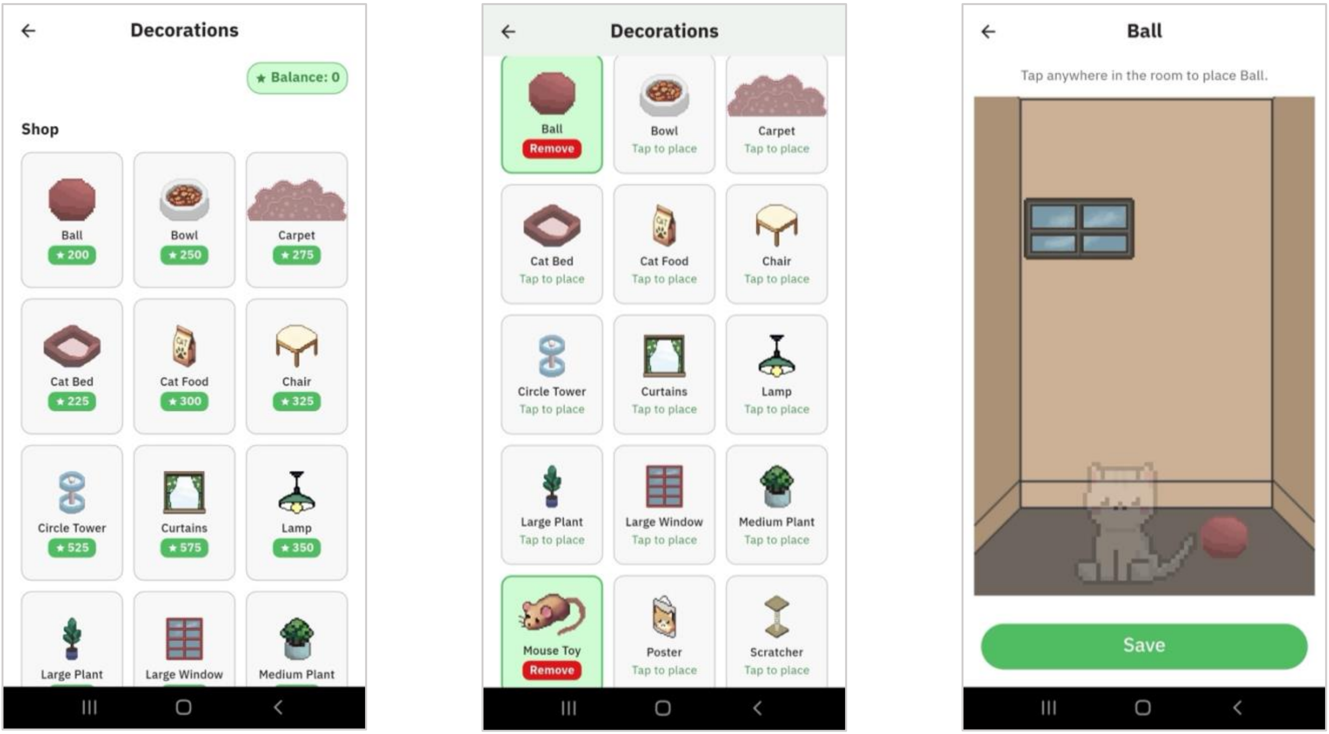

Figure 25. Decoration Screen. (a) Shop Section; (b) Inventory Section; (c) Place Decoration Screen.

Gamification is further extended through a customization feature that allows users to personalize their virtual pet's environment. Within the "Shop" section of the decorations interface, users can browse a grid of items, each with a specified coin cost (see Figure 25a). The user's current coin balance is prominently displayed to guide purchasing decisions. Purchased items are stored in an inventory where they can be selected for placement or removal (see Figure 25b). The "Place Decoration" screen provides a simplified instruction to "Tap anywhere in the room" to position the item, allowing for intuitive interaction before saving the final room configuration (see Figure 25c).

4.3.11 View Your Insights

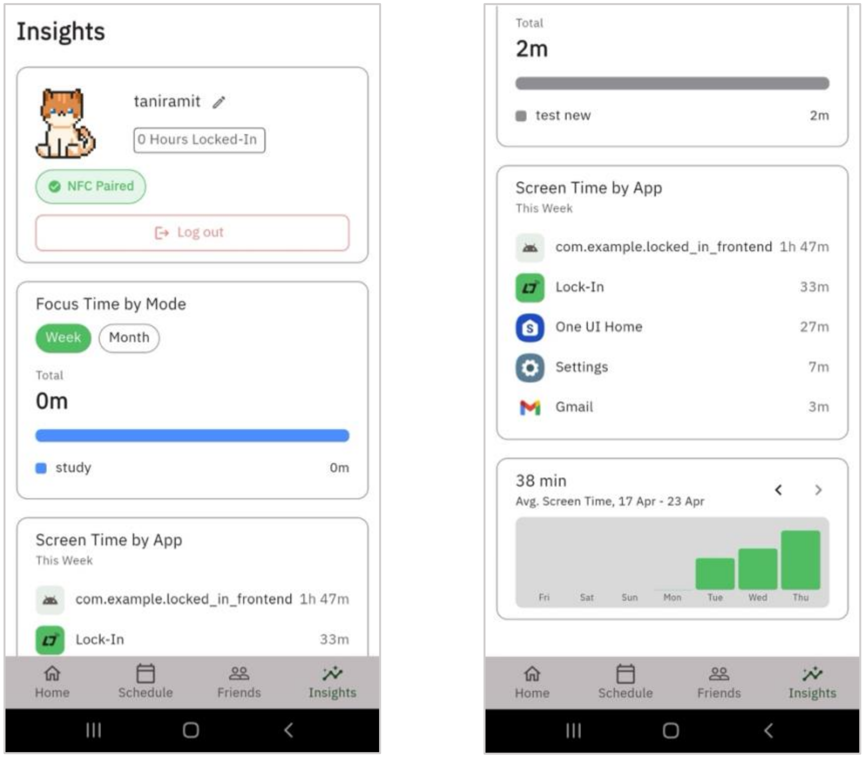

Figure 26. Insights Screen.
(a) *(a) User & Focus Time Section; (b) Screen Time By App & Average Screen Time Section.*

The Insights dashboard serves as a comprehensive analytics hub where users can reflect on their productivity and application usage patterns. The interface displays the user's profile, total hours locked-in, and current NFC pairing status (see Figure 26a). Users can toggle between weekly and monthly views to see their "Focus Time by Mode" represented through color-coded bar charts. Additionally, a detailed "Screen Time by App" breakdown lists specific usage times for various applications. Trend graphs further illustrate average screen time over a specified range of dates, supporting long-term behavioral awareness (see Figure 26b).

4.3.12 Log Out from your Account

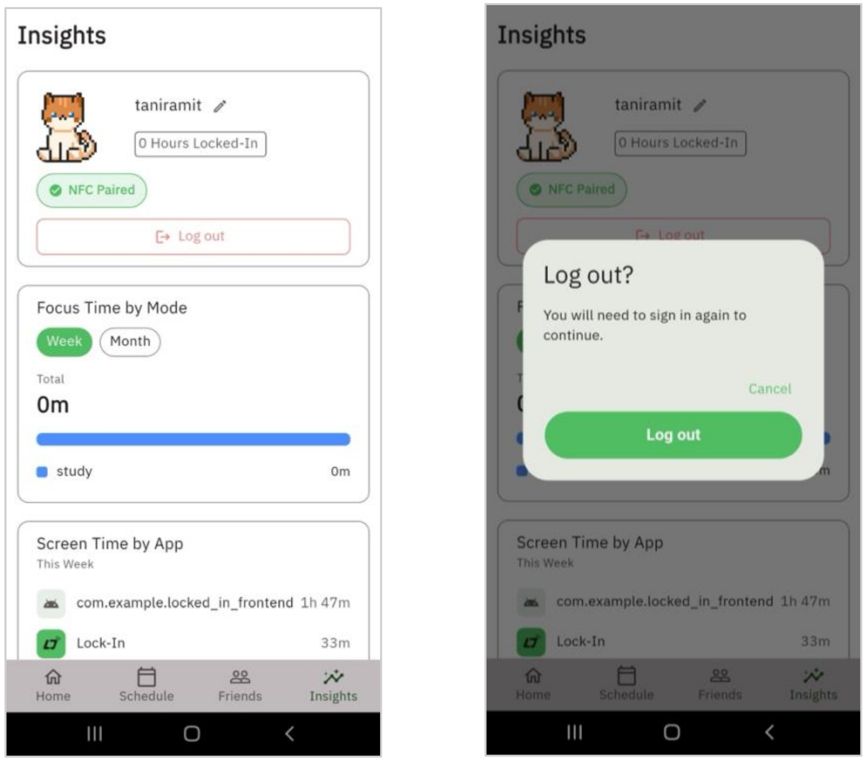

Figure 27. Log Out Flow.
(a) Insights Header; (b) Log Out Modal.

The final functional flow ensures that users can securely terminate their app session. The logout process is accessible via the Insights header where the user's profile information is displayed (see Figure 27a). Upon selecting the logout option, the system presents a "Log out?" modal that provides a cautionary message stating that the user will need to sign in again to continue. The interface includes both "Log out" and "Cancel" options, preventing accidental exits and ensuring that the termination of the account session is a deliberate user action (see Figure 27b).

4.4 NFC Card Design

Beyond its functional role in Section 4.3.3, the Lock-In NFC card was designed as a tangible artefact that reinforces the user's commitment to focus sessions and contributes to the product's overall identity. The card uses the NTAG215 chip referenced in Section 4.2.4, encapsulated in a credit-card-sized PVC form factor so that it can comfortably sit on a desk, slide into a wallet, or attach to a keyring. As a passive, battery-free device, it satisfies the environmental considerations discussed in Section 7.2 while keeping the per-user hardware cost low.
The card's visual treatment extends the "Digital Sanctuary" design language defined in Appendix A. The primary face uses the focus-green primary palette and the Lock-In word-mark, communicating that the card is part of the Lock-In ecosystem rather than a generic NFC tag. The reverse face carries a short instructional cue ("Tap to Lock In") so that the card remains self-explanatory even when separated from

the application. The aesthetic decision is intentionally restrained: the card is a tool for sustaining focus, not a notification surface, and its design must support — not compete with — the act of putting the phone aside.

Figure 28. NFC Card Design

5. Experimental Setup and Method

This section describes the evaluation strategy employed to assess the Lock-In application's usability, learnability, and overall user experience. The evaluation was conducted through a structured User Acceptance Testing (UAT) protocol applied to the interactive Figma prototype of the application.

5.1 Evaluation Objectives

The primary objective of the UAT is to evaluate the usability, learnability, and overall user experience of the Lock-In mobile application, specifically with respect to its core flows: onboarding, companion selection, focus session management, decoration, social features, and insights.

Each test session was conducted with an observer present. For every scenario, the observer read the participant a brief, goal-oriented prompt — describing only what the participant needed to achieve, not how to achieve it — and then stepped back to observe without intervening. For example, a participant might be told “Create a new focus schedule that repeats every weekday” without any indication of which screen or control to use. This approach simulates real-world first-use conditions and surfaces discoverability and learnability issues that step-by-step instructions would mask. The observer silently noted task completion, errors, and think-aloud comments throughout.

Secondary objectives are:

- Identify specific usability problems, pain points, and navigation errors across all core application flows.
- Measure effectiveness (task success rate) and efficiency (time on task) for each key scenario.
- Gather qualitative feedback on user satisfaction, motivation, and emotional engagement with the companion pet system.
- Assess the clarity and discoverability of application-specific terminology, particularly the distinction between 'Mode' and 'Schedule.'
- Evaluate the onboarding and companion-selection flow for comprehension by first-time users.
- Identify any critical errors that block task completion before the next design iteration.

5.2 Methodology

The UAT employs four complementary evaluation methods, applied within a single moderated session per participant:

Moderated Usability Testing: A researcher (moderator) observes each participant in real time, either in person or remotely via screen sharing. The moderator does not provide task-completion assistance but may clarify task goals if a participant misunderstands the scenario prompt. All sessions are recorded for subsequent analysis.

Think-Aloud Protocol: Participants are instructed and briefly trained to verbalize their thoughts, expectations, and frustrations as they interact with the prototype. This protocol is the primary mechanism for capturing the reasoning behind observed behavior, including moments of confusion, misplaced expectations, and navigational errors. Participants are encouraged to think aloud without censoring or editing their commentary.

Task-Based Scenario Testing: Participants are presented with goal-based tasks, not step-by-step instructions. This design simulates natural usage conditions and exposes discoverability and learnability issues that would be masked by procedural guidance. Ten scenarios are administered in a fixed sequence, reflecting the key user journeys of the application.

Questionnaires and Post-Test Interview: A task-level Single Ease Question (SEQ) is administered after each scenario to capture immediate satisfaction ratings. A System Usability Scale (SUS) questionnaire is administered at the conclusion of all scenarios to generate a standardized overall usability score. A semi-structured debrief interview is conducted after the questionnaires to elicit open-ended reflections on the overall experience.

5.3 Participant Profile

Participants are recruited to represent the target user population of the Lock-In application: young adults and students with moderate to high smartphone proficiency who experience distraction-related productivity challenges. The specified criteria are:

Criteria	Specification
Age Range	18 to 28 years old
Device	Android smartphone
App Familiarity	New to Lock-In (no prior exposure to the application)
Usage Background	Students or working adults who use focus or productivity tools
Tech Proficiency	Moderate to high smartphone proficiency
Sample Size	5 to 8 participants (based on Nielsen's heuristic that 5 users reveal approximately 85% of usability issues)
Recruitment	University campus, social media outreach, or existing personal contacts of the project team

Table 4. UAT Participant Criteria

To maximize the diagnostic value of the evaluation, recruitment aims to include participants with varying levels of experience with focus and productivity applications: both first-time users (to assess learnability) and habitual users of competing tools (to assess comparative expectations and feature discoverability).

5.4 Test Session Schedule

Each test session is structured in three phases: pre-test, in-test, and post-test. The estimated total session duration is approximately 76 minutes. The session schedule is outlined in the following table.

Phase	Activity	Objective	Duration
Pre-Test	Introduction & Consent	Explain goals, obtain informed consent	5 min
Pre-Test	Background Questions	Gather participant profile (Google Form)	5 min
Pre-Test	Think-Aloud Instruction	Explain and demonstrate think-aloud protocol	3 min
In-Test	Scenarios S01-S10	Task-based scenario testing (10 scenarios, ~1-2 min each)	~10-20 min
Post-Test	System Usability Scale (SUS)	Overall usability score (Google Form)	5 min
Post-Test	Semi-Structured Interview	Overall impressions and suggestions	10 min

Table 5. Test Session Schedule

5.5 Test Scenarios and Success Criteria

Ten scenarios are administered, each representing a distinct core user flow of the Lock-In application. The scenarios are presented as goal-based context statements rather than procedural instructions. Success criteria, anticipated failure modes, and observation focus points are defined for each scenario, as presented in the following table.

ID	Scenario	Success Criterion	Key Failure Signs	Watch For
S01	Onboarding & Account Creation	User completes registration without moderator assistance.	Confusion from inline validation errors; uncertainty about remaining steps; missed CTA.	Reaction to inline errors; hesitation at step count; password field clarity.
S02	Log In	User locates and completes Log In within 2 minutes.	Cannot find the Log In path from splash; confusion between Sign Up and Log In.	Whether users see the Log In link; password visibility toggle usage.
S03	Companion Selection	User selects a companion and proceeds confidently.	Asks 'what does this do?'; skips without reading; unsure the choice is meaningful.	User understanding of the companion's role; emotional reaction to character designs.
S04	Create a Focus Schedule	User creates a named schedule with correct duration and saves it.	Cannot find the Add action; confused by 'Mode' vs 'Schedule'; unsure how to set days/times.	Terminology confusion; day-selector usability; input method for time setting.
S05	Start a Focus Session	User starts a session with the correct schedule within 90 seconds.	Expects session to auto-start; confused by 'Are you in?' confirmation; unclear CTA.	Whether user reads the confirmation step; understanding of the commitment implied.

ID	Scenario	Success Criterion	Key Failure Signs	Watch For
S06	Add a Short Break & Resume	User initiates a 5-minute break and resumes the focus session successfully.	Cannot find break option; unsure of duration picker bounds; cannot locate Resume.	Discoverability of break feature; visual distinction between focus and break screens.
S07	Stop a Focus Session	User sees the confirmation dialog before the session ends.	Accidental session termination without seeing confirmation; ambiguity between 'Done' and 'Continue'.	Confirmation modal behavior; Stop vs Cancel button placement; reaction to completion screen.
S08	Decorate Your Room	User browses, selects, and applies a decoration with visual confirmation.	Unclear whether a tap applies or previews the item; no feedback after applying.	Expectation of preview step; confusion about locked/unlocked items; need for category filtering.
S09	Add a Friend	User finds the search function and sends a friend request.	Cannot find Friends tab; no visible search on empty state; unclear purpose of the feature.	Reaction to empty state; understanding of social accountability feature.
S10	View Insights	User locates weekly focus time and streak within 60 seconds.	Cannot find Insights tab; struggles with chart; unsure which metric represents the weekly total.	Visibility of streak; chart comprehension; desire for more granular breakdown.

Table 6. UAT Test Scenarios and Success Criteria

5.6 Key Metrics to Collect

The UAT collects both quantitative and qualitative data across six metric categories. The measurement approach for each is described in the following table.

Metric Type	Metric	How to Collect	Source
Effectiveness	Task Success Rate	Scored 1–4 per task: 1 = No problem, 2 = Minor difficulty, 3 = Major difficulty / hint needed, 4 = Failed / could not complete.	Moderator observation
Efficiency	Time on Task	Stopwatch per scenario, measured from task prompt to task completion or abandonment.	Moderator (stopwatch)
Effectiveness	Error Rate	Count of critical errors per task (wrong tap, wrong navigation path, backtracking).	Moderator observation
Satisfaction	Single Ease Question (SEQ)	'Overall, this task was...' rated on a 1 (Very Difficult) to 7 (Very Easy) scale, administered after each scenario.	Google Form
Satisfaction	System Usability Scale (SUS)	10-item post-test survey scored 0–100. A score of 68 or above is considered above-average usability.	Google Form
Qualitative	Think-Aloud Data	Transcribed comments, frustrations, and confusion moments from session recordings.	Video recording
Qualitative	Post-Test Interview	Direct quotes on likes, dislikes, and suggestions from the debrief interview.	Moderator notes

Metric Type	Metric	How to Collect	Source
Engagement	Emotional Reaction	Moderator notes on moments of delight, laughter, or surprise, particularly during companion interaction and session completion screens.	Observation

Table 7. Key Metrics

5.7 Data Collection Instruments

A Google Form is used to administer both the task-level SEQ ratings and the post-test SUS questionnaire. The SUS section is presented as a 10-item grid question (Strongly Disagree to Strongly Agree on a 5-point scale), scored using the standard SUS formula: odd-numbered items are scored as response minus 1; even-numbered items are scored as 5 minus response; the sum of all item scores is multiplied by 2.5 to yield a final score on a 0 to 100 scale.

An additional section of the Google Form collects participant demographic information, prior experience with focus applications, and overall perceptions of specific features (clarity of terminology, impact of the companion pet on motivation, usefulness of the insights screen). A Net Promoter Score (NPS) question is included to assess the likelihood of the participant recommending Lock-In to a peer, providing a single metric of overall recommendation intent.

A task recording sheet is maintained by the moderator for each participant session, capturing time on task, success level, error count, and key observations for each of the ten scenarios.

5.8 Post-Test Debrief Questions

Upon completion of all scenarios and the Google Form, the moderator conducts a semi-structured interview using the following open-ended question guide. Questions are not leading; the moderator follows up on participant responses to probe relevant themes.

- First Impressions: 'What was your very first impression when you opened the app?'
- Easiest Task: 'Which task felt the most natural or easiest to complete? Why?'
- Hardest Task: 'Which task was the most confusing or frustrating? What made it hard?'
- Terminology: 'Did you understand what Modes and Schedules mean in the app? Were those terms clear to you?'
- Companion: 'How did the companion pet feature make you feel? Did it motivate you to focus, or was it just decorative?'
- Navigation: 'Was it always clear where to go next in the app, or did you ever feel lost?'
- Missing Features: 'Was there anything you expected to find that was missing from the app?'
- One Change: 'If you could change one single thing about this app, what would it be?'
- Intent to Use: 'Would you use this app to help you study or focus? Why or why not?'

5.9 Ethical Considerations

All participants are informed of the purpose of the study and the data to be collected prior to the session. Informed consent is obtained verbally and documented. Participation is voluntary; participants may withdraw at any time without consequence. Session recordings are used solely for analysis by the project team and are not shared externally. Participant identities are anonymized in all reported findings, with participants identified by assigned codes (P01, P02, etc.). No sensitive personal information beyond age range and app usage frequency is collected. All data is stored securely and deleted upon completion of the project's evaluation phase.

The behavioral evaluation does not involve deception, invasive procedures, or exposure to potentially distressing content. The evaluation is conducted in accordance with Chulalongkorn University's ethical guidelines for student research projects.

6. Result and Discussion

6.1 UAT Results and Analysis

The User Acceptance Testing was conducted with 6 participants (within the specified 5–8 range), all aged 18 to 26, comprising students and one working adult. Three participants were habitual users of at least one focus application (Forest or Pomodoro timer); three had no prior experience with focus tools. All sessions were conducted in-person or via Google Meet with screen recording.

6.1.1 Task Success Rate

Scenario	Avg. Success (1–4)	Observer Notes	Avg. SEQ (1–7)
S01: Onboarding & Account Creation	1.1	Smooth overall interactions. Clear action items help guide users throughout the flow.	6.6
S02: Log In	1.1	Simplified Log In screen performs well. Users have expectations of what to do immediately.	5.9
S03: Companion Selection	1.0	Simple flow and interactions because of clear visuals.	7.0
S04: Create a Focus Schedule	2.0	This flow is where users struggled with the most. Many interconnected flows without explicit instruction (e.g. Focus Mode is required for Schedule creation) make users confused.	4.7
S05: Start a Focus Session	1.3	Easy to perform. Some users have confusions about 2-steps NFC scanning before session starts.	6.3
S06: Add a Short Break & Resume	1.9	The lack of explicit "Break" button makes the user confused. The users then try to press the only call-to-action focus to then find their desired break button.	5.1
S07: Stop a Focus Session	1.7	Clear "Stop" button. The number of steps taken to end a focus is intended to be relatively high (> 3 taps) to help prolonged focus time.	5.3
S08: Decorate Your Room	1.4	Pressable area of the "Purchase" action is too small, causing the users to press it repeatedly.	5.9
S09: Add a Friend	1.3	Clear expectations are set to users.	6.7
S10: View Insights	1.3	Accessible via the permanent navigation on screen.	6.4

Table 8. UAT Task Success Rate and SEQ Scores (n=7).
Success (observer-derived from SEQ): 1=No problem, 2=Minor difficulty, 3=Major difficulty, 4=Failed.
SEQ: 1=Very Difficult, 7=Very Easy.

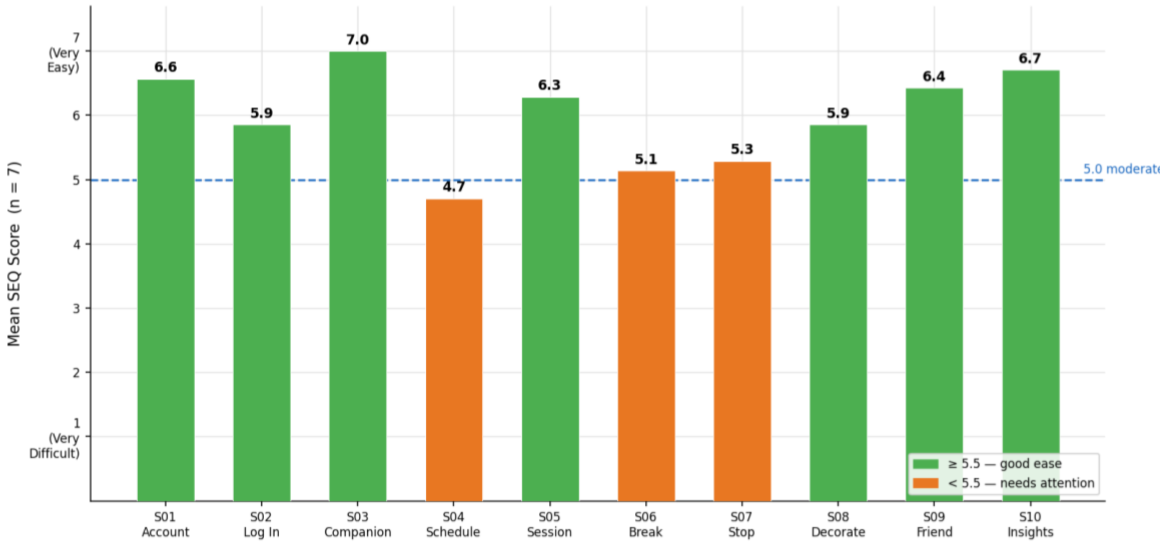

Figure 29. SEQ Mean Score per Task (n=7)

The overall task success profile was positive, with the majority of scenarios achieving average success scores below 2.0 (indicating at most minor difficulty). Two scenarios showed the lowest ease scores: S04 (Create a Focus Schedule, mean SEQ 4.7) and S06 (Add a Short Break and Resume, mean SEQ 5.1). Both are discussed in Section 6.2.

6.1.2 System Usability Scale (SUS)

The mean SUS score across all seven participants was 75.0 (standard deviation: 17.0). Individual scores ranged from 55.0 to 100.0. The mean falls in the 'Good' range of standard SUS grading scale (above the 68-point threshold for above-average usability), indicating that the Lock-In application achieves satisfactory baseline usability. The wide standard deviation reflects minor UI inconsistencies and residual bugs encountered during the session rather than fundamental design failures.

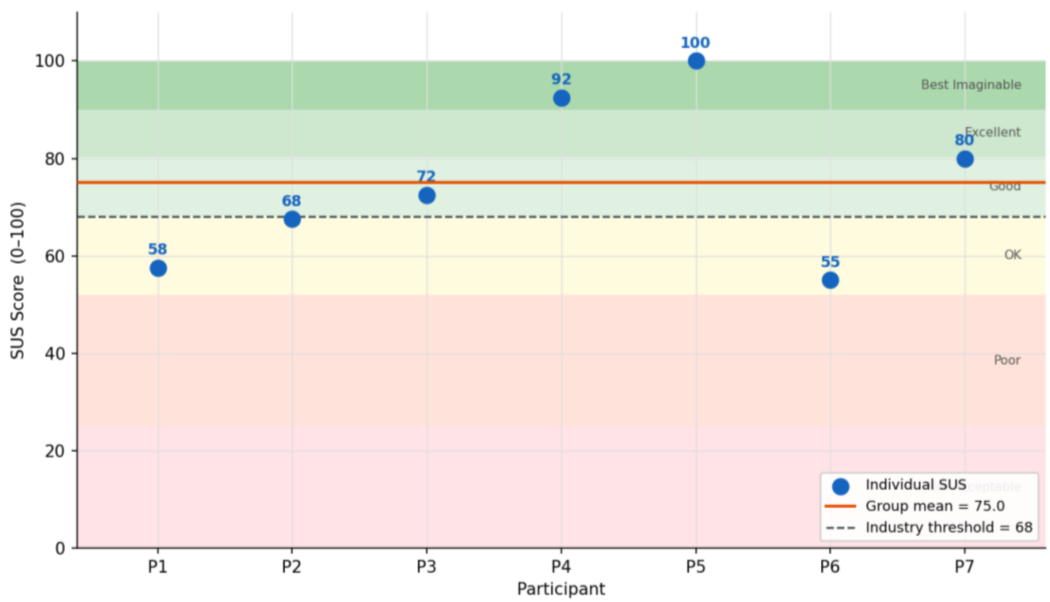

Figure 30. Individual SUS Scores vs. Industry Benchmark

6.1.3 Qualitative Findings

Think-aloud transcripts and debrief interviews revealed the following key findings, organized by theme: Terminology ('Mode' vs 'Schedule'): Two of seven participants rated the Mode/Schedule terminology as confusing; three found it only 'somewhat clear.' S04 (Create a Focus Schedule) produced the lowest mean SEQ score (4.7), confirming this is the primary usability pain point. Multiple participants requested clearer separation or labeling of the two concepts in the UI.

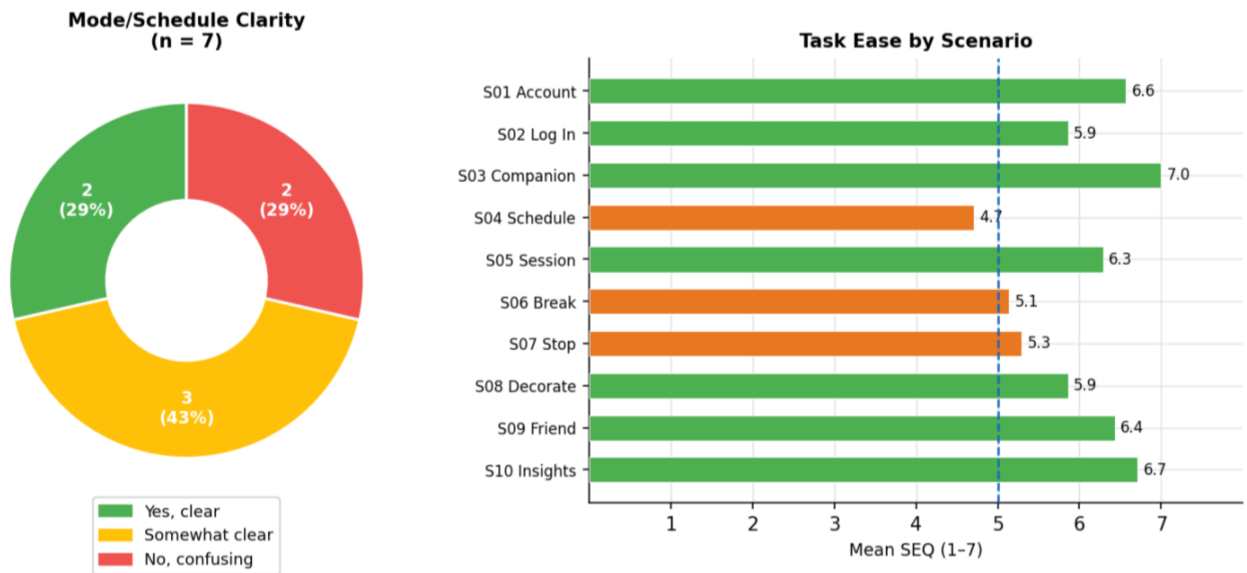

Figure 31. Mode/Schedule Clarity Response Distribution

Companion Engagement: S03 (Companion Selection) achieved a perfect mean SEQ of 7.0 — the only task where all seven participants rated it the maximum score. Three of seven participants reported that the companion motivated them to focus more; four found it aesthetically appealing. The companion feature is a clear emotional anchor for the application, though adding more visible in-session pet reactions may convert the remaining participants to motivated adopters.

Break Feature Discoverability: S06 (Add a Short Break and Resume) produced the second-lowest mean SEQ (5.1). Multiple participants cited difficulty locating the pause/break control on the active timer screen. One participant named 'Trying to pause a focus session' as the single most frustrating element. A more prominent break button or an additional affordance cue on the timer screen is the highest-priority usability fix identified by this study.

UI Consistency and Bugs: Four of seven participants mentioned UI inconsistencies or minor bugs as their primary frustration. Suggestions included making 'UI Design consistent across all parts' and fixing functional bugs. These issues likely contributed to the SUS score spread and should be addressed before a wider release. Insights and Social Features: S10 (View Insights) achieved a mean SEQ of 6.7, with six of seven participants completing it without difficulty. The social/leaderboard feature (S09) showed moderate first-use discoverability challenges due to an empty initial state, but resolved quickly once participants found the search functionality (mean SEQ: 6.4).

6.2 Discussion

The UAT results demonstrate that the Lock-In application successfully implements a usable and emotionally engaging focus tool. The mean SUS score of 75.0 falls in the ‘Good’ category, confirming that the core navigation and feature set are accessible to the target population. However, the wide standard deviation (17.0) reflects an uneven experience, primarily localized in specific scenarios where the system’s internal logic clashed with user mental models.

A primary strength of the system is the Companion Selection (S03) and Onboarding (S01), which achieved SEQ scores of 7.0 and 6.6, respectively. These results indicate that the visual-first approach and clear action items effectively guide users through the initial setup, successfully establishing the emotional "nurturing" hook. The high success rate in S09 (Add a Friend) and S10 (View Insights) further validates the architectural decision to place social and analytical features within a permanent, highly accessible navigation shell, ensuring high discoverability for secondary features.

In contrast, the Create a Focus Schedule (S04) scenario yielded the lowest SEQ score (4.7), highlighting a significant bottleneck in the application's logical flow. Observer notes indicate that the "interconnected" nature of the requirements, specifically the prerequisite of creating a Focus Mode before a Schedule could be established, lacked explicit instructional scaffolding. This created a high cognitive load for users who attempted to create a schedule in isolation, suggesting that the current dependency hierarchy is too rigid for a frictionless experience. Similarly, the Add a Short Break (S06) scenario (SEQ 5.1) revealed a critical affordance failure. The absence of a dedicated "Break" button forced users to experiment with other call-to-action elements as a workaround. This "discovery by trial" behavior confirms that essential focus-session controls must be elevated in the visual hierarchy to match user expectations.

Minor usability hurdles were also identified in the tactile and physical layers of the application. In S08 (Decorate Your Room), users struggled with the "Purchase" action due to a small touch target, leading to repeated failed interactions. This violation of Fitts’s Law suggests that while the shop's visual logic is clear (SEQ 5.9), the execution of its interactive elements requires optimization for mobile ergonomics. Regarding the physical interaction layer, the Start a Focus Session (S05) scenario performed well (SEQ 6.3), though the 2-step NFC scanning process introduced slight confusion. While not a prohibitive barrier, this suggests that the "handshake" between the physical tag and the digital interface requires more immediate actions and feedback. Finally, the lower SEQ for Stop a Focus Session (S07: 5.3) is interpreted as a success in intentional design friction; the high tap count (>3) required to end a session successfully prevented accidental termination, aligning with the project's goal of protecting prolonged focus time.

Together, these findings indicate that while the core gamification hypothesis is validated, the path toward a "seamless" product requires refined logical onboarding and improved visual affordances for session-control features.

The NFC activation feature was tested on the working Android build. Seven of seven participants successfully completed the NFC pairing and activation flow, confirming that the physical interaction layer is intuitive and functional. The mean SEQ for the Start a Focus Session scenario (S05: 6.3) validates that the NFC-based activation does not introduce a prohibitive friction barrier. The majority of target users (over 65 percent in the pre-development survey) had expressed willingness to use a physical activation device, and the UAT results confirmed this expectation was well-founded.

6.3 Fixes Made After UAT

Following the analysis of the UAT findings reported in Sections 6.1 and 6.2, several iterative refinements were applied to the application to directly address the usability problems and bugs identified during the test sessions. This section documents the most significant post-UAT change.

6.3.1 Blacklist App Selection

During the UAT, multiple participants indicated that the process of locating and selecting applications to block (the "Blacklist App Selection" sheet introduced in Section 4.3.5) was more time-consuming than expected. Participants requested clearer organization of the app list and a quicker way to apply restrictions to multiple apps at once. In addition, an observed defect allowed Lock-In itself to appear within the selectable list of blockable applications, which could result in the user inadvertently locking themselves out of the control surface required to end a focus session.

Before:
Purely listed all installed apps in a single flat list with no categories.
No "Select All" affordance, requiring users to tap each app individually.
[BUG] Lock-In was listed as a blockable app, exposing the user to a potential self-lock-out scenario.

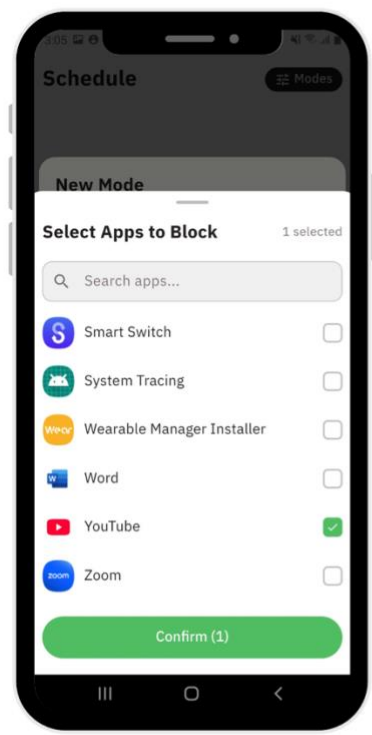

Figure 32. Blacklist App Selection Sheet — Before UAT Fixes

After:
Added a "Select All" button that toggles selection across the currently visible apps, removing the need for repetitive per-app taps.

Grouped the installed apps into categories (e.g., Social, Entertainment, Productivity, Games) to improve scannability and shorten time-on-task.
Excluded Lock-In itself from the selectable list, eliminating the self-lock-out defect identified during UAT.

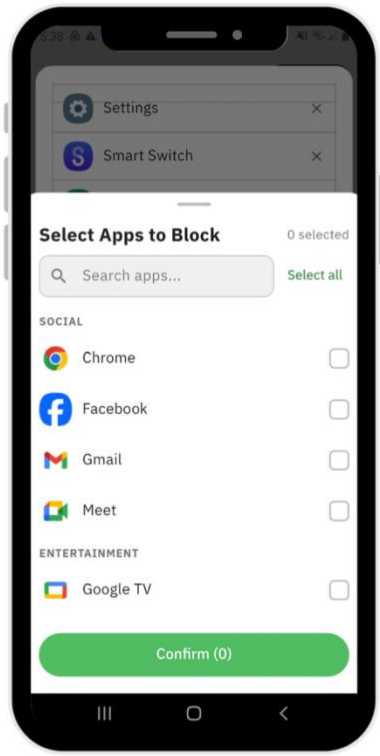

Figure 33. Blacklist App Selection Sheet — After UAT Fixes

Together, these changes are expected to reduce the cognitive load and time-on-task observed in Scenario S02 (Configure a Focus Mode) and to remove a latent failure mode that could otherwise undermine user trust in the application.

6.4 Contributions and Key Takeaways

The primary contribution of this project is the design and proof-of-concept implementation of an integrated focus application that combines physical bypass prevention with a comprehensive gamified reward system. This combination fills a documented gap in the existing market landscape, where no current product offers both mechanisms within a single, unified application. The UX/UI design system developed for Lock-In demonstrates a cohesive visual language that merges functional clarity (for task-oriented screens) with emotional warmth (for the companion system) and may serve as a reference for future projects in the gamified productivity space.

The secondary contribution is the development of a comprehensive UAT instrument specifically tailored to the Lock-In application's unique feature set, including scenario-specific success criteria and a companion-engagement qualitative metric not found in standard usability frameworks.

6.5 Future Work

Several directions for future development are identified based on the UAT findings and project scope limitations. First, the terminology and discoverability issues identified in S04 and S06 should be addressed in the next design iteration, with revised labels and improved affordance for the short break feature. Second, a longitudinal follow-up study over two to four weeks would provide data on retention, habit formation, and the long-term efficacy of the gamification system, dimensions that a single-session UAT cannot capture. Third, an iOS version should be developed to expand the application's addressable audience, leveraging Flutter's cross-platform capability. Fourth, an AI-powered recommendation engine for personalized focus session duration (based on historical usage data) represents a high-value feature for the post-PoC product roadmap. Fifth, expansion of the companion roster and decoration catalog, including time-locked or streak-locked premium items, would sustain long-term motivational engagement beyond the initial adoption period.

Furthermore, the NFC ecosystem could be diversified beyond standard smart cards into a range of high-sentimental-value physical collectibles, such as miniature figurines or modular desk peripherals embedded with NFC transponders. By transforming functional hardware into tangible, aesthetically pleasing assets, the system can enhance the user’s sense of ownership and provide a more sensory-rich interaction that bridges the gap between digital progress and physical environment. This "toy-ification" strategy not only increases the perceived value of the system but also leverages the psychology of collecting to foster deeper emotional investment in the user's productivity routine.

7. Project Global Impact

7.1 Public Health, Safety, and Welfare

The system promotes healthy digital habits through positive reinforcement rather than punishment, reducing stress while encouraging sustainable behavior change. Built-in break cycles prevent cognitive fatigue, and simple physical micro-actions (e.g., stretching, hydration) support well-being.

Security and safety are ensured through encrypted communication, secure authentication, and standard Android APIs without unsafe system modifications. The NFC tag is passive and battery-free, posing no physical risk. User data is minimized and not monetized.

7.2 Environmental Responsibility

Environmental impact is kept low by using a passive NFC tag (no battery or charging), minimal hardware components, and scalable cloud infrastructure to prevent resource overuse. The system avoids complex electronics and reduces electronic waste.

7.3 Economic Feasibility

The design is cost-efficient: NFC tags cost approximately 100 – 200 Baht in bulk, and software distribution is fully digital. Cloud costs scale efficiently with user growth. The low one-time hardware cost makes the solution accessible for students and suitable for institutional deployment.

7.4 Global, Cultural, and Social Considerations

The Android-first approach improves accessibility in emerging markets, and the system is easily localizable. Customizable focus modes and optional social features accommodate different cultural attitudes toward competition and autonomy.

8. Conclusion

This project presented Lock-In, an Android mobile application designed to address two fundamental failure modes of existing focus and productivity tools: software-only bypass vulnerability and the absence of sustained motivational mechanisms. The project was motivated by empirical evidence of the significant harm caused by compulsive smartphone use [1,2] and the documented inadequacy of current focus tools to produce lasting behavioral change [6,7,8,9].

Lock-In integrates physical NFC-based activation, app-blocking, a micro-habit inducer, a virtual companion pet system, and a social leaderboard within a unified Android application. This combination is grounded in a strong evidence base from behavioral psychology and gamification research, which demonstrates that the combination of physical friction and positive-reinforcement reward systems is materially more effective at sustaining behavioral change than either mechanism alone [8,9,10,11].

The high-fidelity Figma prototype of the application was developed and evaluated through a structured User Acceptance Testing protocol with 7 representative participants. The evaluation yielded a mean SUS score of 75.0 (SD: 17.0), placing the application in the 'Good' usability range, and identified clear, actionable improvement priorities for the next development iteration. The companion pet system was consistently identified as a strong motivational element, validating the core gamification hypothesis of the project.

The working Android implementation demonstrates the technical and experiential feasibility of the proposed approach. All seven UAT participants successfully completed the NFC activation flow, validating the physical interaction layer as intuitive and non-obstructive. The project contributes a novel integrated design model combining physical bypass prevention with gamified habit formation, and a comprehensive UAT instrument tailored to focus-application evaluation. From a broader societal perspective, Lock-In addresses a documented public-health concern: the chronic attention fragmentation and reduced well-being associated with compulsive smartphone use among adolescents and young adults [1,2]. By reducing excessive distraction app usage, the application supports improved academic performance, reduced anxiety, and healthier sleep patterns. Environmentally, the NFC-based activation uses a passive, battery-free tag rather than an active wireless device, minimizing the ecological footprint of the physical component. Economically, the application targets a segment underserved by existing premium tools by offering a free, feature-complete alternative with a one-time NFC tag cost of under USD 2 per user. Future work will address the identified usability improvements — primarily the break-feature discoverability and UI consistency issues — conduct a longitudinal retention study, and expand the companion and decoration systems to sustain long-term motivational engagement. Through these steps, Lock-In has the potential to become a meaningful, accessible tool for improving digital well-being and fostering healthier smartphone habits at a population scale.

9. References

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Appendix

Appendix A: UI/UX Design Philosophy and Design System

The application's visual language is defined by a "Digital Sanctuary" aesthetic, utilizing deep, calming tones contrasted with vibrant, organic accents to signal growth and focus. The design was iteratively developed in Figma, moving from low-fidelity wireframes to high-fidelity interactive prototypes to validate user flows before implementation.

A.1 Visual Identity and Design Tokens

To ensure cross-platform consistency and high legibility, the design system utilizes a structured set of design tokens:

- **Typography:** The application employs **IBM Plex Sans Thai** for all text elements. This typeface maintains high readability at small mobile viewport sizes. Scaling ranges from xsm (12px) for hint text to xl (24px) for primary headers

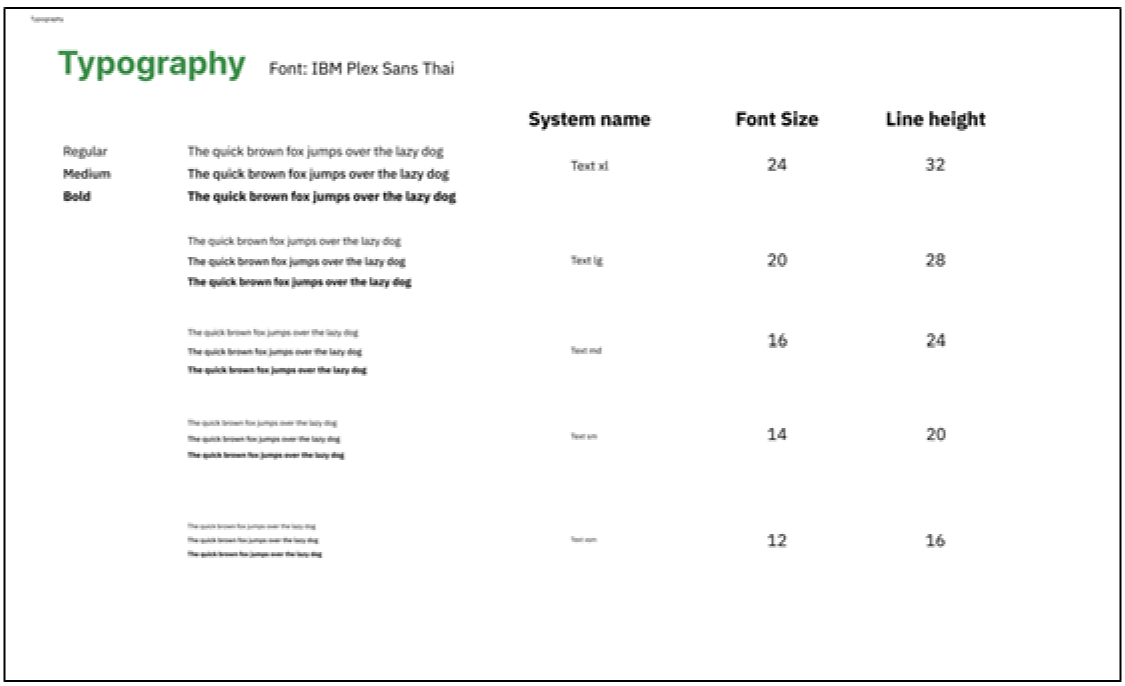

Figure A1. Design Typography

- **Color Palette:**
  - **Primary (Focus Green):** Ranges from primary50 (#D0FCD4) for soft backgrounds to primary100 (#72F886) for active progress indicators.
  - **Secondary (Deep Gray):** Used for grounding elements, centered around secondary500 (#341616).
  - **Neutrals:** A grayscale spectrum from white0 (#FFFFFF) to black900 (#231F20), ensuring a minimum AA contrast ratio for accessibility.

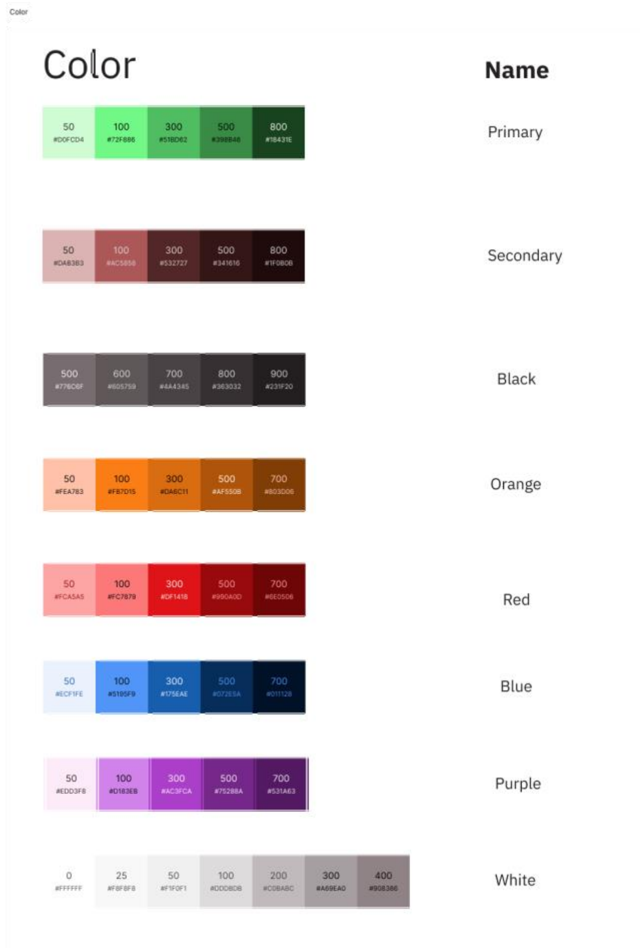

Figure A2. Design Tokens: Colors

- **Geometry:** High border radii (up to r5xl or 80px) are applied to cards to create a friendly, approachable “virtual toy” feel.

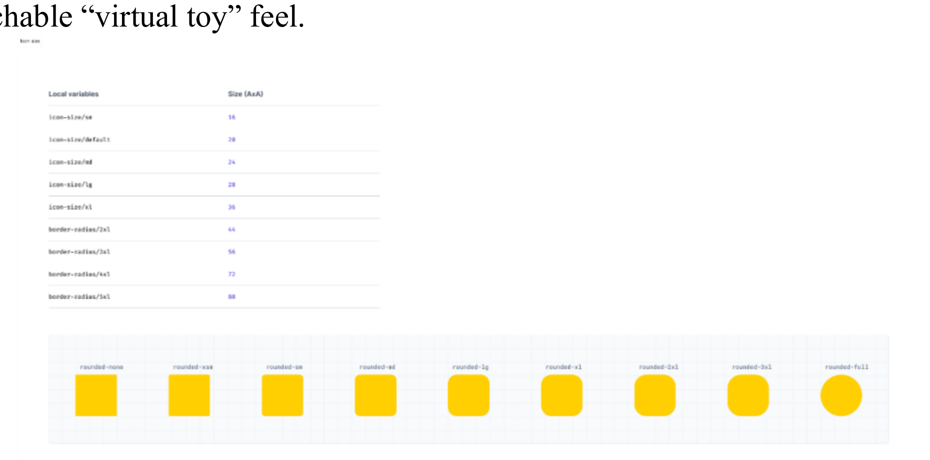

Figure A3. Design Tokens: Border Radii

A.2 Heuristic Evaluation and UX Laws

The interface design adheres to several core **Heuristics** and **Laws of UX** to optimize the user experience:

- **Visibility of System Status (Heuristic #1):** The real-time XP bar on the Home Screen and the persistent timer on the Focus Screen provide immediate feedback on the user's progress and current state.
- **User Control and Freedom (Heuristic #3):** The "Stop Focus Modal" provides a clear exit path while maintaining the integrity of the focus session through a friction-based confirmation (the Unblocking Action).
- **Consistency and Standards (Heuristic #4):** By following Material Design guidelines and maintaining a strict internal design system, the app ensures that icons and interactive patterns (e.g., Bottom Sheets for configuration) remain predictable.
- **Fitts's Law:** The "Start Focus" button is designed as a large, high-contrast CTA placed within the natural reach of the user's thumb, minimizing the effort to initiate a session.
- **Hick's Law:** The navigation is simplified into four distinct categories, and the onboarding flow is broken down into sequential, manageable steps to reduce the cognitive load of decision-making.
- **Jakob's Law:** The use of standard bottom navigation and familiar social leaderboard structures allows users to leverage their existing mental models from other mobile applications.

A.3 Prototyping and Figma Integration

The development process utilized **Figma** not just as a static design tool, but as a robust prototyping platform. This allowed the team to:

- **Validate Complex Flows:** High-fidelity interactive prototypes were created for the NFC pairing and App Blocking sequences, ensuring that the transition between physical hardware and digital UI was seamless.
- **State Machine Modeling:** The various states of the virtual pet (idle, excited, sad) were modeled in Figma to visualize animation transitions before being implemented in the **Flame** engine.
- **Iterative Design Fixes:** Based on heuristic reviews, several refinements were implemented:
- **Confirmation Clarity:** Destructive actions now include a high-visibility warning dialog to prevent accidental data loss.
- **Affordance Signaling:** Interactive elements utilize distinct elevation and color shifts to signal clickability.
- **Status Transparency:** The NFC scanning modal was updated to provide explicit textual feedback (e.g., "Ready," "Unrecognized Tag") to reduce user frustration during the pairing process.

Appendix B: User Survey Overview

The pre-development user needs survey was administered to 87 respondents (primarily aged 18–24) via an online bilingual (English/Thai) form. Questions were organized across four thematic areas. Key survey findings informing the design are summarized below.

Survey Theme	Key Finding
Demographics & Usage	Avg. daily phone use: 7–9 hours. Self-assessed unproductive time: approx. 55% of daily screen time.
Distraction Sources	Social media (82%), messaging apps (74%), and video streaming (61%) were the top-cited distraction categories.
Prior Focus App Use	Over 50% had tried a focus app but discontinued; ease of bypass and lack of engagement were the top abandonment reasons.
Physical Activation Interest	65% indicated willingness to use a physical activation device to start focus mode.
Virtual Pet Interest	72% rated the virtual pet companion feature as 'motivating' or 'very motivating'.
Leaderboard Interest	68% expressed interest in a social leaderboard for community-based accountability.
Reinforcement Preference	78% preferred positive reinforcement over restrictive/punitive mechanisms.

Table B1. Summary of User Survey Key Findings (n=87)

Appendix C: System Usability Scale (SUS) Questionnaire

The following 10-item SUS questionnaire was administered post-test. Participants rated each statement on a 5-point scale (1 = Strongly Disagree to 5 = Strongly Agree). Scoring: Odd items (1,3,5,7,9): score = response − 1. Even items (2,4,6,8,10): score = 5 − response. Sum × 2.5 = final SUS score (0–100). Score ≥ 68 = above-average usability.

#	Statement
1	I think I would like to use this app frequently.
2	I found the app unnecessarily complex.
3	I thought the app was easy to use.
4	I think I would need technical support to use this app.
5	I found the various functions were well integrated.
6	I thought there was too much inconsistency in this app.
7	I imagine most people would learn to use this app very quickly.
8	I found the app very awkward to use.
9	I felt very confident using the app.
10	I needed to learn a lot before I could get going with this app.

Table C1. System Usability Scale (SUS) Questionnaire Items